

Scoping the Lasting Effects of *The Lord of the Rings*

Report to The New Zealand Film Commission

April 2002

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Preface (I)

In August 2001, NZIER were retained to undertake a study which would “*enable the New Zealand Film Commission to make an informed assessment of the effects of the production of ‘The Lord of the Rings’ (three feature films) on the domestic film industry and on selected areas of the New Zealand economy*”.

This report is called a scoping study and is speculative, to a degree — because there is considerably more work yet to be done on the trilogy and little comparable experience regarding extent and duration of effects.

It discusses the drivers of the New Zealand film industry, where possible presenting evidence on the likely size of the lasting economic effects.

Its scope is aided by the degree to which thoughtful insights have been provided to the reviewers by many industry participants. We are grateful to those who have made such contributions to the information available.

This scoping study will represent a platform on which further analysis could be undertaken. The initial conclusions could then be tested over time as the trilogy project is completed and more information becomes available.

We have noted in the report that in order to identify actual effects, as distinct from the projections and scenarios reported here, it will be important to have a reliable and standardised monitoring system in place to capture both qualitative and quantitative information.

This report was prepared at NZIER by John Yeabsley and Ian Duncan, and reviewed by Alex Sundakov. We are grateful to Liz Hodgson for her major editorial and layout contribution.

Preface (2)

I am delighted to see this first public study of the long-term contribution to New Zealand made by *The Lord of the Rings*.

In discussing the project with the Associate Minister for Arts, Culture and Heritage, Hon Judith Tizard, I expressed the hope that aspects of the project be documented so that we might use this experience to better understand how to grow New Zealand's film industry. My own development as a filmmaker was strongly assisted by the New Zealand Government through the NZ Film Commission. I am happy to be able to make a contribution in return.

I believed that this was so important that I sought the co-operation of New Line, that the data to underpin this study could be made available for analysis. I am grateful to New Line for their co-operation and generosity.

I am committed to international filmmaking driven creatively from New Zealand and I look forward to many other New Zealand filmmakers making their films here using the best talent New Zealand and the international filmmaking community have to offer for the benefit of New Zealand as a whole both economically and culturally.

Peter Jackson

Preface (3)

The brief for this scoping study arose from a dialogue between Peter Jackson, the Associate Minister of Arts, Culture and Heritage, Judith Tizard, and the New Zealand Film Commission. We are grateful to Peter Jackson, Barrie Osborne, Janine Abery, Elena Azuola, Deborah Fox, Three Foot Six Limited, New Line Cinema, Ian Macfarlane and all the other people who committed time and information to the project.

The report stresses the importance of creative entrepreneurship in the development of the New Zealand film industry. It reinforces the emphasis of the Film Commission on working with the new generation of film entrepreneurs to improve the creative product and increase their international connections. This new focus commenced in January at Cinemart in Rotterdam and will see the Commission taking a more structured approach to supporting filmmakers in overseas markets.

This scoping study recommends further work on monitoring the qualitative and quantitative elements of film projects. We will be talking with our industry partners to see how this can best be done.

The Lord of the Rings demonstrates clearly the benefits of a creatively-driven film industry. *The Lord of the Rings* originated in New Zealand and was pre-produced, produced, filmed and post-produced here. It leaves a unique and lasting footprint. It leaves significant intellectual property and human capital gains. It has changed the way the film world views New Zealand, our capabilities and the risk of doing business here. It has given New Zealand a stunning new profile in our key tourism markets.

The fact that three quarters of all expenditure on *The Lord of the Rings* to date has been spent in New Zealand, on the work of New Zealanders, is testament not only to Peter Jackson's exceptional talent and commitment to this place but also to the pool of talent which has been developed here.

The Film Commission is extremely proud to have supported Peter Jackson in the making of his first four New Zealand films. The core mission of the Film Commission is to seek out talented New Zealand filmmakers and to give them an environment within which to work and develop. We will continue to encourage the development of creative entrepreneurs because it is the creation of projects within New Zealand that is the best way of ensuring that more major film projects will be made here.

Significantly, this study shows that *The Lord of the Rings* project has changed the aspirations of our filmmakers. It has extended the limit of their dreaming. It is expanding the possibilities of what they can achieve and this in turn will bring enormous benefits to New Zealand's visibility in the world. Bring on those dreams.

Ruth Harley

Chief Executive, The New Zealand Film Commission

SUMMARY

TERMS OF REFERENCE

The objective of this Scoping Report is “to enable the New Zealand Film Commission to make an informed assessment of the effects of the production of *The Lord of the Rings* on the domestic film industry and on selected areas of the New Zealand economy”.

The brief was to focus on the economic effects, transitory and (most importantly) lasting, of the production of *The Lord of the Rings* in New Zealand. The terms of reference require a focus on employment, infrastructure, and support services

We note that generalisations about lasting effects are speculative, and must be treated with care. However, we have drawn a set of initial conclusions which can be tested over time as more information becomes available.

We find that the three films comprising *The Lord of the Rings* have had unique effects on New Zealand, extending beyond the scale of the already significant transitional effects.

The trilogy will leave a unique ‘footprint’ for New Zealand when its production is over.

CRITICAL ELEMENTS

We find that one of the key lasting effects should be a change in the probability of major feature films being made in New Zealand. This will be influenced by local creative individuals with international credibility. It will also be affected by the appeal of local resources, value for price, technical backup, suitable scenery, helpful regulations, and critically by the ability to control cost and risk during the shoot.

We find that New Zealand's perceived competitiveness as a production base has been significantly enhanced as a result of production of the trilogy here.

We also find that the making of *The Lord of the Rings* may have an important effect on films reflecting local themes and culture, in terms of increasing activity and markets.

We find that the New Zealand production-skill base and capacity has been broadened and deepened, and we discuss likely lasting effects on New Zealand's creative reputation, talent development, production capacity, as well as people-based effects and creative entrepreneurial effects.

We find that a principal effect of *The Lord of the Rings* (which has been unprecedented in its magnitude) has been to lift industry capability and ability to new levels, especially in terms of managing large and complex production processes, solving problems in complicated technical and creative areas, and enhancing networks with skilled New Zealand technical and production teams.

We find indications that tourism spin-offs could be significant, and that there has been useful exposure.

We analyse the significance for New Zealand of a growing proportion of film financing and production activity resulting from projects developed in New Zealand, in comparison with projects developed overseas.

In this context, we record the fact that, from mid-1998 to early March 2002, about 74 per cent of the trilogy's total production and post-production costs, and the same sort of proportion of all labour costs were spent in New Zealand.

TRANSITIONAL EFFECTS

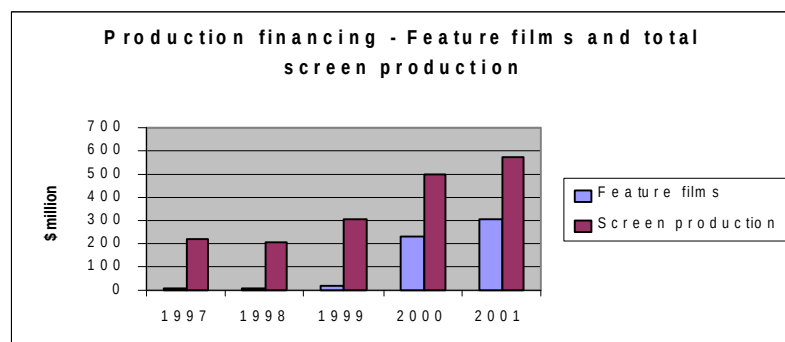
With the release of *The Fellowship of the Ring*, and the continuing post-production of the next two features, transitional effects include the following highlights (all in NZ dollars):

- \$352.7million expenditure by the production company in New Zealand (to March 2002). The above New Zealand expenditure includes:
 - labour costs of \$187.7million
 - digital effects costs of \$99million
 - miniatures and creature costs of \$36.5million
 - location costs of \$31.3million
 - construction costs of \$25.1million
 - transportation costs of \$12million
- This level of expenditure produced peak period employment of around 1500 people per week (This number does not include any day labour or extras).
- It is about 3,200 person years employment of New Zealand tax residents for the four years from 1997 to 2001.
- Around 5000 vendors were used, most of them in New Zealand.
- Expenditure will continue as the next two films are prepared for release.

GROWTH OF THE SCREEN PRODUCTION INDUSTRY

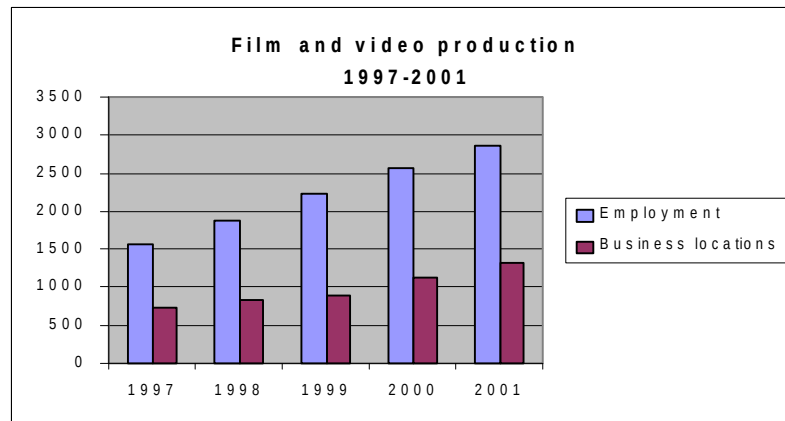
Including the effects of *The Lord of the Rings*, comparative data for the New Zealand screen production industry shows that investment in New Zealand feature films grew from \$16million in the year ending March 31, 1999, to \$308million in the year ending March 31, 2001. In the same period, employment in film and video production rose from 2,240 to 2,860.

Figure 1: Production financing



Source: Colmar Brunton Survey, 2001

Figure 2: Film and video production



Source: Statistics New Zealand

LASTING EFFECTS

We expect that there will be seven main categories of lasting effects resulting from the production of *The Lord of the Rings* trilogy.

These are:

1. Raising the international profile of New Zealand film-writing, the New Zealand production and post-production industry, and talented individuals in these sectors;
2. Upskilling the New Zealand screen production industry at both technical and management levels;
3. Establishing a foundation of New Zealand-based creative entrepreneurship, centred on the New Zealand film industry;
4. Encouraging a significant attitudinal change amongst New Zealand writers, producers and directors towards larger projects, and the more determined pursuit of investors;
5. Broadening and deepening film-related infrastructure and contributing to a more user-friendly regulatory environment;
6. Enhancing Brand New Zealand, for example by opening additional New Zealand based tourism;
7. The potential for spin-off industries such as merchandising, and miniatures.

We provide more detail of these likely lasting effects.

Taking them together, the potential for success by New Zealand talent as creative forces in the film industry has significantly been enhanced. New Zealand is no longer just a scenery-based location. Any future international productions in New Zealand should be denser in local content than was previously likely and the need to import crew is substantially reduced.

We specify limits to the weight that can be placed on these conclusions.

EMPLOYMENT

It seems probable that *The Lord of the Rings* experience will result in significantly more film production activity in New Zealand and will increase capacity utilisation.

INTERNATIONAL PROFILE

The Lord of the Rings has enhanced international perceptions of New Zealand's capability. The perceived "risk" of New Zealand as a "distant" and chancy production base has thereby been reduced.

ECONOMIC SCENARIOS

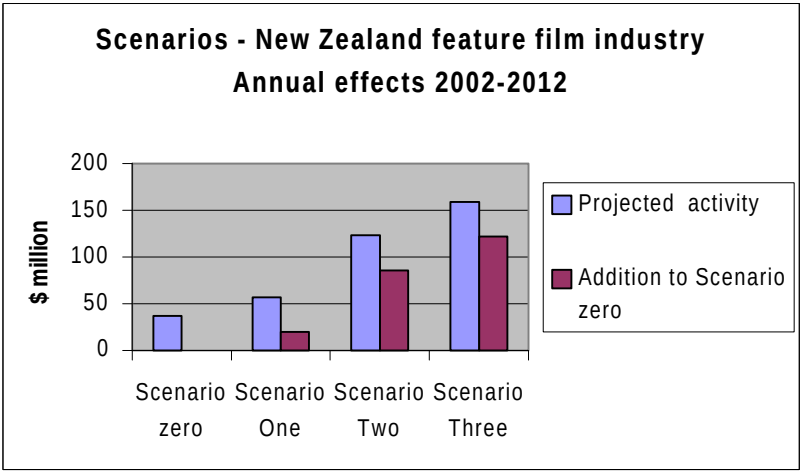
Our assessment of the possible future of the New Zealand film industry resulting from the effects of Peter Jackson's *The Lord of the Rings* trilogy has been made in comparison with assumptions about the industry in the absence of that project. As a model for "business as usual without *The Lord of the Rings*" does not exist, we have created "scenario zero" as a standard of comparison for three scenarios of possible future film industry activity over the next ten years or so.

Scenario One, steady growth from the existing value base, shows an average increase in New Zealand film activity of the order of \$20million, relative to Scenario Zero.

Scenario Two, with an increasing New Zealand share of internationally mobile films, results in an average increase of New Zealand film activity of the order of \$85million a year.

Scenario Three, showing a takeoff in entrepreneurial activity, brings about an increase in New Zealand film activity averaging of the order of \$120million a year.

Figure 3: Economic scenarios



Source: NZIER

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PART I INTRODUCTION

These sections provide a brief review of the task, and how we approached it. They also include some initial background on the film industry. This is at a simplified level, but is intended to provide a basis for the analytical structures that we build on in later sections.

I. OUR BRIEF

I.1 The task

“The objective of this study is to enable the New Zealand Film Commission to make an informed assessment of the effects of the production of *The Lord of the Rings* on the domestic film industry and on selected areas of the New Zealand economy.”¹

Our *Terms of Reference* states that the study should focus on:

1. Employment;
2. Infrastructure, including the extent to which the production built on existing infrastructure and developed new infrastructure relating to film production in New Zealand; and
3. Support services and the development of their capacity.

In other words, the focus of the research is on supply responses in New Zealand — prior to, coincident with, and after the production process. Our focus in this study is on two aspects of supply response, with the second of these being the critical and unique part.

- *Transitory supply responses*, with no real enduring changes. For example, a local coffee shop sells more — or perhaps modifies its range to include espressos — while the film is being produced, and possibly even makes temporary alterations in its operation (a take-out service?), but does not make any long-term change in the way it functions. After the film is made, and the cast and crew depart for new projects, the coffee shop is back to where it was before the film project commenced, having had a good run in the meantime.
- *Lasting ‘capacity effects’* in the infrastructure of the film industry, the people engaged, and supporting activities, and possibly more widespread. Again, given the size of the project being considered, activity locally will have inevitably expanded and then waned. But, more importantly, making *The Lord of the Rings* here will have left lasting imprints on the quality and range of New Zealand’s ‘soft’ and ‘hard’ infrastructure, and on its international profile as a film production location.

I.2 The approach

The brief is specifically focused on the economic effects, transitory and lasting, of *The Lord Of The Rings*. In order to address these questions we needed to build a picture of:

- Trends in the New Zealand film industry prior to *The Lord of the Rings*.
- The effects of *The Lord of the Rings* on local and film industry resources, especially people and production facilities.
- Factors considered by international studios, directors and producers in choosing production locations.

A considerable part of the research effort went into primary information collection — basically extensive interviews with a range of key people — to understand their involvement with *The Lord of the Rings*, how it has affected them and the various production houses.

These interviews were the source of factual information on contributions at the pre-production, production, and post-production stages. They also provided more general insights into ‘how the industry works’ in New Zealand, and on how the industry operates internationally, and on the linkages between these.

¹ Contract for Consultancy Services, Annex A, August 2001

We have interspersed italicised extracts from the notes of these interviews throughout the report. These are not attributed, and are not necessarily precisely verbatim, but reflect the sentiments, and the way these were expressed. They are used selectively to support or illustrate points in the way we were brought into the picture.

Interviews, together with various international references on the economics of the film industry, and other sources of wider information, provided the main elements of the economic framework we use for our analysis.

1.3 Period studied

As required by our brief, the study relates to the period beginning with the commencement of pre-production to post-production for film one. This corresponds with the period from mid-1998 to early 2002 for which data has been presented.

1.4 Report structure

The report is organised as follows.

Part I: Introduction

Section 1: Our brief

Section 2: Introduction to film

Part II: The New Zealand film industry

Section 3: Film industry trends in New Zealand to 1998

Part III: Effects of *The Lord of the Rings*

Section 4: The New Zealand film industry from 1998

Section 5: Creative entrepreneurs

Section 6: New Zealand film production components

Section 7: Film industry capacity in New Zealand

Section 8: Production capacity

Section 9: Internationally mobile films

Section 10: Other spin-offs

Section 11: Initial conclusions

Section 12: References

Appendix A: Quantification of film prospects

Appendix B: International film industry

Appendix C: Platform for further work

Appendix D: Thanks

Appendix E: Above- and below-the-line

Appendix F: The three parts of creating a movie: pre-production, production, and post-production

Appendix: G Estimating exposure worth

2. INTRODUCTION TO FILM

2.1 Defining the industry

In analysing the effects of the *The Lord of the Rings* on the local ‘industry’ we need to be clear about what we mean by the ‘industry,’ and where it ‘fits’ in a broader economic sense.

Film is a medium for projecting moving images. The film industry can be characterised as a stand-alone industry, or as part of a much broader information, communications and technology sector. As per Magder (1993, p.245),

“The cinema still retains its allure and its status as a premier cultural institution — for purposes of marketing and publicity if nothing else — but to speak of a film industry divorced from the television (or audio - visual) production industry is no longer a very useful analytical distinction.”

In New Zealand, films are seen as part of a broader screen production industry. We note though, that demand patterns, risks, and cost structures in feature films are likely to be quite different from those say, of television production, where there will typically be an assured market for the projects undertaken. Films and television material may include feature films, short films, tele-movies, one-offs, series and serials. Subject matter and style/genre can cover a wide spectrum.^{2 3}

2.2 Influences on New Zealand as a production location

Films have a major creative element, and are important mirrors of society. But feature film making is a business, in which, like any business, investors aim to make profits commensurate with the risks involved. Our analysis of the effects of *The Lord of the Rings* has to be within an appropriate context. That is an international one, in which most of the risk capital for production is generated or assembled in a few countries, but in which many countries now offer highly developed film production infrastructure.

We are interested in this because one of the key lasting effects of *The Lord of the Rings* should be a change in the probability of international feature films being shot in New Zealand.

Two main influences on the choice of New Zealand as a production location are:

1. Proactive or ‘above-the-line’ effects — The intervention of local, internationally credible, creative individuals, whose reputation is extremely high. Their involvement in the project must be sufficient to encourage the international funders (who will inevitably be fundamental to a significant film) to fall in line with the advice they are offered that New Zealand is a suitable location; and
2. Reactive or ‘below-the line’ effects — These essentially determine the ‘competitiveness’ of New Zealand as a film production location. This assessment will depend on the particulars of the project concerned, but will be likely to stem from the combination of local resources, value for the price, appropriate back up in a range of technical areas, varied and suitable scenery, helpful and compliant regulations and government agencies, and other relevant matters, including critically, the ability to control cost and creative risk during the shooting process.

² Refer Productivity Commission (2000) *Broadcasting Inquiry Report* no.11, March 2000, Canberra (pp.147–154).

³ Refer Lois S Gray, and Ronald L Seeber (1996) *Under the stars — essays on labor relations in arts and entertainment*. Cornell University Press.

This is not to ignore the lasting effects of this project on New Zealand-made films reflecting local themes and culture. *The Lord of the Rings* may have an important effect on the markets for, and level and type of activity in such film-making. This will involve all the complicated interactions between workers and facilities that already takes place. But it is realistic to assume that the major share of film investments, for films with significant size budgets, will continue to be sourced from investors overseas.

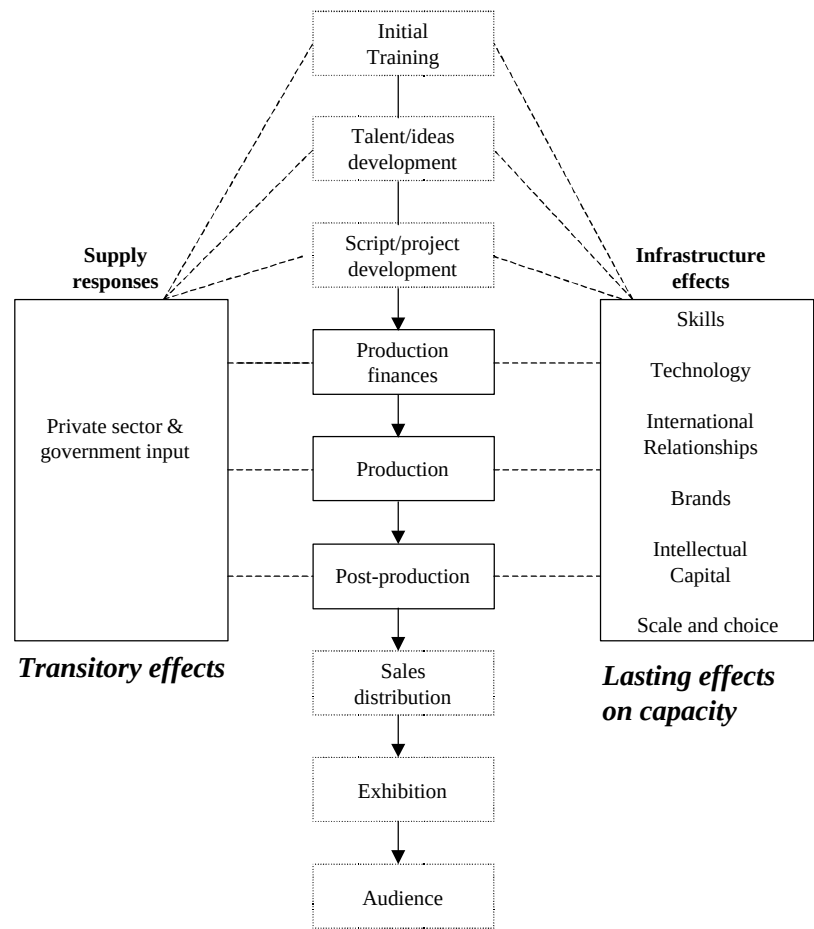
2.3 Focus of our research

For research and reporting purposes we needed to have a clear understanding of the processes involved in the making of *The Lord of the Rings* and, in particular, the supply linkages and likely effects of the film on the capacity of suppliers and film industry-related infrastructure.

In other words, what are the ‘dynamics’ of the local supply responses and infrastructure effects — the size of these responses (e.g. employment, investment etc.), their nature (i.e. qualitative information), and duration?

Figure 4 provides a broad picture of the context. The value chain flows vertically down the centre of the figure from ‘script/project development’ through to the ‘audience’. Our attention, in this report, has been mainly on the boxes with solid boundaries in the centre of the diagram, associated with the actual production process. We are also interested in the initiation process that lies above these in the diagram, ‘talent/ideas development’ and ‘script/project development’. But there are also important feedback effects from exhibitions and from audiences.

Figure 4: Film industry value chain and supply responses



Source: NZIER adaptation of New Zealand Film Commission material

In these terms, the transitory and lasting effects of *The Lord of the Rings* production activities can be located around the central chain of processes, and in the box on the right side of the diagram. We have focused in this report mainly on the effects indicated on the right-hand side of this diagram, and on what they mean for the ‘capacity’ of the New Zealand film industry in the round.

2.4 Stylised model

The aim here is to set out a simplified, but representative picture of the environment in which talent selection and backing, film investment and location, decisions are being made internationally. We have reduced a complex and sophisticated business to some generalised features for the purpose of the analysis that follows.

Key considerations are:

- Films are strongly ‘front-loaded’ investments, in which considerable sums have to be spent, on pre-production, production, and post-production, as well as on distribution, and promotion, usually many months and sometimes even years, before any revenues flow in from paying cinema audiences.⁴
- Large budget films are usually taken to be those with budgets of \$US25 million or more.⁵ *The Lord of the Rings* is a ‘monster’ project, encompassing three films, with a total production budget, said by commentators, to be about \$US300 million.⁶
- Although potential audiences are huge, the financial success of any one film is never predictable — each film is a hand-made ‘one-off,’ often with no assured market. These can be highly successful low budget movies which return many times their cost, or blockbusters that burn large holes in investors’ pockets.
- Potentially, feature films can tap an expanding range of collateral revenue sources including cinema audiences, television viewers, and the video and DVD market. The latter groups — including cinema audiences outside the US — have steadily expanded in recent years as a proportion of the total return. There is also potential for well publicised, or niche oriented, films to have spin-offs in the merchandising area.

Spending on films falls into two broad categories, ‘above-the-line’ and ‘below-the-line.’ (Refer Appendix E) The first of these refers to the writer, executive producers, director, and other ‘management’ staff, as well as the stars and supporting cast members.

‘Below the line’ are most of the inputs into the production process, for example, special effects, sets, costumes, make-up, stunts, photography and so on. To date, this is where most of New Zealand’s skills and resources have developed.

Management of one of the production facilities that worked on *The Lord of the Rings* told us that:

“Despite New Zealand’s much lower cost structure than the US or other possible sources of special effects, The Lord of the Rings project had to be finely priced to be sure that we would be awarded the work. Technical ability, together with business skills, have been necessary to survive in a narrow, price-sensitive, and unpredictable marketplace.”

⁴ Again this is a generalisation. Sales of distribution deals and product placement or merchandising rights, for example, can significantly reduce the degree of front loading and dependence on variable box office results.

⁵ In the US currently, ‘large’ may indicate budgets of \$US60 million or more.

⁶ Refer for example, Sunday Star Times, October 7, 2001.

2.5 International film industry context

Appendix B covers some generic economic features of the international film industry that are pertinent to this assessment.

Main topics covered there are:

- Risk management and flexible specialisation (B 1)
- Labour market conditions (B 2)
- The evolution of the international film industry (B 3)
- Internationally mobile projects (B 4)
- US reports (B5).

2.6 Risk management in the film production sector

Risk is a combination of the probability of some occurrence, and the consequences of that occurrence.

Management of risk is a feature of all businesses, and a key consideration and influence in the film industry. This applies to people and individuals ‘above-the-line’ and ‘below-the-line.’

“Making films is a business and there will always be trade-offs between quality and costs. But, in my view, and for all serious projects, creative decisions should be made before the financial constraints kick in. New Zealand is not the cheapest filming location, and it would be wrong to market itself as such, in the wake of a project as exciting as The Lord of the Rings. When budgets come first, projects end up with the tail wagging the dog.”

For investors, an important part of the risk derives from the Hollywood ‘no one knows anything’ syndrome. The box office success or otherwise of films is notoriously unpredictable. The optimum strategy for investors in this setting is to have a ‘portfolio’ of film investments, with the inevitable risk spread across a range of producers, stars, genres etc.

For those in the production sector, an important element of risk is the fluctuating pattern of demand they are likely to face. They might have a series of projects in quick succession, or several months downtime between substantial projects.

There are several relatively standard business and theoretical responses to this type of risk profile:

- **Flexible specialisation**, as discussed in Appendix B. This protects economic viability, first, through the ability to offer upward flexibility in capacity in a market with lumpy demand; and second, through the ability to shed overheads quickly to financially manage downtime.
- **Investment**, in ongoing marketing and networking to increase the probability of being in the loop for new projects, and thus lowering the potential for necessary rest periods.
- **Diversification**, into related fields, such as merchandising, electronic game design commercial production, etc.

These are broad generalisations, and precise strategies adopted will vary within and across the different components of the production sector.

PART II THE NEW ZEALAND FILM INDUSTRY

One ultimate purpose of this research is to make an assessment of the extent to which *The Lord of the Rings* has or will enhance the New Zealand film industry, in some lasting way. What are the prospects for the industry now compared with what it would have been had *The Lord of the Rings* not been made in New Zealand?

In order to address this issue, we have to understand the way that the structure of the film industry has been evolving internationally, and what this means for future trends in New Zealand. Changing market conditions influence the size and shape of this industry, as they do any industry. The following are brief ‘scene setting’ overviews rather than attempts at comprehensive coverage.

They relate to a New Zealand film production sector which is involved with both New Zealand-developed and funded projects, as well as with projects, which are principally developed and/or funded offshore.

3. FILM INDUSTRY TRENDS IN NEW ZEALAND TO 1998

3.1 Overview

With one or two notable exceptions, the New Zealand film industry is focused on the production side, so most people employed in it actually work for service suppliers. These firms may also service the needs of the television and advertising industries, for example. Potentially, films shot locally may be originated by a variety of different organisations, including:

- Locally based production companies, which may be one-offs set up for the particular project;
- Co-production teams, which are joint projects between local and foreign partners who share creative control and finance;
- Foreign production companies, which are based overseas but who produce films and programmes in New Zealand;
- Television broadcast companies; and
- Educational institutions, and community groups.⁷

3.2 New Zealand film industry trends to 1998

This section provides a brief statistical profile of film industry trends up to 1998, the year in which pre-production commenced on *The Lord of the Rings*.

3.2.1 Employment and activity units

Here we confront the problem of coverage - which activities should be included as part of the film industry and which should not? The effects of *The Lord of the Rings* are likely to be concentrated in the first of the industry categories i.e. film and video production, but we also show data for related categories to provide a broader view.

Table 1: Employment in film related industries

Full-time equivalents

February	1994	1995	1996	1997	1998
Film and video production	1312	1607	1497	1570	1876
Film and video distribution	145	171	209	215	191
Motion picture exhibition	544	748	825	1016	947
Sound recording studios	102	132	149	177	195
Total	2103	2658	2680	2978	3209

Source: Statistics New Zealand *Business Activity*

⁷ Productivity Commission, 2000, p.148.

Note that film production at any given time is likely to include a large number of short-term independent contractors. To provide numbers which can be used to sum across industries or compare between them, Statistics New Zealand uses full-time equivalent persons engaged.⁸

3.2.2 Infrastructure and support services

Overview

Infrastructure for film production comprises:

1. Film-specific infrastructure: knowledge and skills (*soft* infrastructure), buildings, equipment, film processing, specific telecommunication links (hard infrastructure);
2. General supporting infrastructure: for example, construction, communications, accommodation, transport, catering, repair services, power, general supplies, and so on;
3. Government services and regulation administration.

There is some advantage in having permanent facilities for some parts of the film production process, as long as they can be adapted for use in differing projects. Specialised movie production and processing equipment frequently has a short economic life, perhaps as little as one to two years for modern electronic gear.

“From a continuing capacity viewpoint, this steep fall-off in the value gradient emphasises the rapidity with which a hardware-based position in the market can erode. From a financial perspective, it also emphasises the care needed in timing and sequencing of large equipment purchases, and the requirement for significant cash flow-projects to fund these purchases.

As is the case for specialist equipment, capital outlays on production facilities need to be linked as closely as possible to funded projects, and investors need to structure their organisations to leverage down to a sustainable minimum to allow for probable downtime. This approach relies, too, on the owners reinvesting surpluses from previous projects.”

Central and local government relations should also be included (either as part of the infrastructure, or support services). In addition to organisations such as the New Zealand Film Commission, Film New Zealand, Investment New Zealand, Industry New Zealand, with a mandate to support the industry, central and local government has a general regulatory role.

Examples in central government include:

- Immigration processes, which influence visa or residency applications by offshore personnel;
- OSH, ACC and other labour-related laws;
- Income tax rules, including particularly those specific codes applying to the self-employed;
- DOC management of access to the public estate.

At the local level are consent processes related to the building of sets, land use for filming etc., which can be critical in facilitating or obstructing the logistics of a film project.

Evolution to 1998

Development of film-specific infrastructure is not a steady process, but a reflection of the previous experience. Thus it relates to timing, size, and effects of films that have already been made in New Zealand. While most of the feature films made in New Zealand over the last decade or so have been relatively small to medium sized, they have contributed importantly to the platform of soft-infrastructure, in particular.

⁸ Full-time equivalent persons engaged (FTE) equals the sum of full-time employees and working proprietors plus half the part-time employees and working proprietors.

PART III: EFFECTS OF *THE LORD OF THE RINGS*

Gauging the effects of *The Lord of the Rings* needs to be done by reference to the relevant base case, that is, the New Zealand film industry in the absence of that project. For example, if we want to measure the employment effects of *The Lord of the Rings*, this could be done directly, or by looking at total employment in the film industry and deducting from it some assumed ongoing level of employment without it.

In this part of the report we set out data on the New Zealand film industry — as far as possible making explicit the *transitory* effects of *The Lord of the Rings* since about 1999.

We also discuss likely *lasting* effects, in terms of New Zealand's creative reputation and linkages, people-based effects, talent development, creative entrepreneurial effects, production capacity, attraction of internationally mobile films, and other spin-offs.

4. THE NEW ZEALAND FILM INDUSTRY FROM 1998

4.1 Data assessment

Our quantitative assessment of *The Lord of the Rings* (in late 2001 and early 2002) is both backward looking and forward looking. Conceptually the first component is relatively straightforward. The second component is inherently speculative and imprecise.

The scope of this project meant we were not able to systematically collect any primary overview data for this assignment. This part of the assessment, set out in more detail in Appendix A draws on the following data categories:

1. Accounting data from Three Foot Six Limited (the production company for *The Lord of the Rings*) covering the period 1998 to early 2002. This covers employment and major categories of expenditure. This can be taken as soundly based and accurate, but the employment data requires some adaptation to put in a form comparable with other economic data.
2. Statistics New Zealand *Business Activity* statistics which set out data for employment and activity units by industry up to February 2001.
3. Colmar Brunton data on employment and expenditure by media type. This is survey based and requires some further analysis to make it comparable with other economic data.
4. Other numbers are used as part of more general descriptions of the film industry but these are not purported to be rigorously collected.
5. International data, on production and expenditure trends for feature films. Although this is not directly from official statistical agencies, for our purposes it can be taken at face value.

4.2 Employment

We have noted that ‘employment’ in the film industry is project-based. During certain peaks, there may be large numbers of people with contracts in the industry, but many of these contracts may be for only a few days or weeks.

The effects of a large project such as *The Lord of the Rings* should be to increase:

- The pool of people in New Zealand with film industry experience;
- Their prospects of getting work here or elsewhere in the industry in the future;
- The utilisation of the existing pool (during the filming or other production work for *The Lord of the Rings*)

*“On the other hand, the ‘churn’ associated with the international business means that local industry could tend to lose institutional memory, and people who would ideally be retained e.g. for head of department roles. Another ‘negative’ coming from the high profile that *The Lord of the Rings* is likely to achieve, is that overseas production houses will ‘cherry pick’ particular crew or creative people, who might otherwise have stayed on the local scene.”*

The Statistics New Zealand data below provides snapshots as at February of each year of full-time equivalents in the industry and gives some impression of the effects of *The Lord of the Rings*. Note that we cannot impute from this the precise net employment effect of *The Lord of the Rings* because we have no way of knowing exactly what would have happened in the absence of that particular film project.

Note also that these figures are at one time each year — measurements at other times might give a different impression of the trend.

4.2.1 Statistics New Zealand

Table 2 shows the number of geographic units (business locations) for film and video production and other film-related industries. It shows strong growth in the former, particularly from 1999.

Table 2: Film-related industries business locations

February	1997	1998	1999	2000	2001
Film and video production	740	831	883	1137	1324
Other film related industries	611	599	601	673	655
Total	1351	1430	1484	1810	1979

Note: These are separate operating units engaged in New Zealand in one, or predominantly one kind of economic activity from a single physical location or base

Source: Statistics New Zealand *Business Activity*

We can also provide a regional breakdown of business locations and employment specific to *film and video production*. The largest absolute and relative increase between 1999 and 2001 was in Wellington, and it would seem to be reasonable to attribute much of this to *The Lord of the Rings*.

Table 3: Film and video production business locations

February	1997	1998	1999	2000	2001
<i>Urban areas</i>					
Auckland	416	482	513	626	693
Wellington	185	186	184	283	387
Rest of NI	49	60	63	69	74
Christchurch	36	39	50	61	64
Dunedin	18	22	20	25	30
Rest of SI	16	17	21	34	30
<i>Rural areas</i>					
	20	25	32	39	46
Total	740	831	883	1,137	1,324

Source: Statistics New Zealand *Business Activity*

Similarly, much of the employment growth in film and video production occurred in Wellington, although Auckland also recorded significant growth.

Table 4: Employment in film and video production

Full-time equivalents

February	1997	1998	1999	2000	2001
<i>Urban areas</i>					
Auckland	920	1,120	1,320	1,290	1,550
Wellington	380	430	470	570	800
Rest of NI	75	110	110	120	110
Christchurch	70	85	100	130	130
Dunedin	50	50	100	130	140
Rest of SI	40	40	75	280	65
<i>Rural areas</i>					
	35	40	55	60	75
Total	1,570	1,880	2,240	2,570	2,860

Source: Statistics New Zealand *Business Activity*

4.2.2 Colmar Brunton data

The eighth in a series of these surveys was published in November 2001. Because of the method used to collect the data, they are subject to various caveats.⁹ However, they provide some additional insights into employment trends over the years in which much of the filming for *The Lord of the Rings* was underway, as well as other production and post-production work.

Table 5: Positions in screen production

Type of employee	1998	1999	1999	2000	2000	2001
Independent contractors or freelancers		6,412		12,760		29,589
Permanent positions:						
Part-time positions		552		535		541
Full-time positions		766		1045		1137
Total positions		7,729		14,340		31,266
Notes:	(1)	Part-time positions are defined as those in which people are employed all year but for less than 20 hours per week.				
	(2)	Full-time positions are defined as those including contracts spanning more than 40 weeks.				

Source: Colmar Brunton (2001), p.19.

4.3 Infrastructure and support services

In Section 4.2.2, we discussed the nature of the infrastructure associated with film production as combining both ‘soft’ (embedded knowledge and skills) and hard (equipment, buildings, communication links).

We suggested three main categories of this hard and soft infrastructure:

1. Film-specific infrastructure.
2. General supporting infrastructure: for example, transport, catering, repair services, power.
3. Government services and regulations.

Because demand is project by project, the critical skill is to be able to be responsive but tightly managed, whatever the size of the project or the style of film.

“In filming the entire Lord of the Rings saga, Peter Jackson was doing something that had never been done before. He was setting out to prove that it was more efficient to film three Hollywood blockbusters simultaneously rather than three separate movies. The enormity of the task meant that up to five separate units were filming at once. The problem was that Peter could only be at one of them, and that’s not the ideal way for keeping your vision intact.

A few months before the cameras started rolling, The Lord of the Rings asked Telecom what they [Telecom] could do for them. Telecom responded with a world-first satellite link-up that allowed Peter to be on set with a video, voice and data link to three or more other locations. While directing the action, he could keep an eye on the work of the other units. It had to be a robust connection too, with military grade optic fibre tough enough to survive the horses’ hooves and Queenstown floods.”

(Interview with Jamie Selkirk, *Unlimited*, December 2001 / January 2002)

Our assessment, based primarily on interviews with specialists in many parts of the production process is that the principal effect of *The Lord of the Rings* has been on the soft infrastructure components. The picture is of the components of the New Zealand industry, including a series of supporting contributors, as having climbed to new levels on an industry ability and capability ladder.

⁹ *Survey of Screen Production in New Zealand 2001*. Prepared by Colmar Brunton for Screen Producers and Directors Association of New Zealand. p.19.

This is especially so in terms of:

- Extensive and sustained experience in managing large and complex production processes, to meet strict budgets and deadlines within flexible and shifting creative demands.
- Problem solving skills across a wide front and on a variety of scales in the complicated technical/creative areas including at Weta Physical, and at Weta Digital.
- Enhanced confidence and networks with the people who support the general capability, via the diverse technical and skill teams that can be pulled together in New Zealand, to produce feature films.

Previous film production in New Zealand (features, and programmes for television) provided the background for some of the steps in this ladder, but there can be no doubt that *The Lord of the Rings* effect has been unprecedented in its magnitude.

Previous investment in 'hard infrastructure' especially by the companies associated with Peter Jackson contributed to the foundation for *The Lord of the Rings*. That in turn, has allowed substantial reinvestment in such facilities (e.g. film studios, production houses, substantial digital processing). Development of this infrastructure base is an important adjunct to *The Lord of the Rings* especially in conjunction with the 'softer' human capital, management development, and entrepreneurial effects, already discussed.

5. CREATIVE ENTREPRENEURS

5.1 Overview

In Section 2.2, we suggested that there were two main influences on the choice of New Zealand as a production location:

- Proactive or ‘above-the-line’ *entrepreneurship*
- Reactive or ‘below-the-line’ *support*.

This is shorthand, and an oversimplified dichotomy. For example, international marketing activity by New Zealand production houses can influence the flow of projects coming here. So we can be proactive in attracting films developed elsewhere for shooting and production here. But the main distinctions are important in considering the possible lasting effects of *The Lord of the Rings*, and determining factors.

We are drawing on the broader economic literature relating to the concept of entrepreneurship¹⁰ to relate to the concept of a process of *alertness to profit opportunities*. In Harper’s words:

“The entrepreneur recognises ... that there is an opportunity waiting to be grabbed.”

This supports the active and vital role that we are concerned with here. A person who is seen in the industry as able to identify such opportunities, is able to take others (including financiers) with him/her, as the project is developed. It is an active and positive role – seizing initiatives and making things happen. By its nature this is a risky process and not one able to be systematised. It often accompanies or is integrated with the role of being creative; not just spotting opportunities, but actually inventing them. To cover both aspects (the opportunism and the creative/talent side) of the concept, we use the term ‘creative sponsor’ here.

This activity is in sharp contrast to the support role, that we discuss here as ‘below-the-line.’ Such activity can far more be easily turned into a routine, and while involving some action (and even initiative taking), the spark for the projects must come from someone else.

Turning back to *The Lord of the Rings*, the effects on the New Zealand film industry have been broadly:

- To potentially develop a capacity for New Zealanders to be creative sponsors for the development and control of major international projects to be filmed and completed in New Zealand through enhanced entrepreneurship potential;
- To broaden and deepen the New Zealand-based production skill base; capacity in general, and the perceived competitiveness of New Zealand as a production base through enhanced support capacity.

5.2 New Zealand developed film projects

Until very recently, prospects for the New Zealand film industry were thought to depend on the small amount of New Zealand Film Commission investment available, and some rather random distribution of projects developed overseas, but filmed and produced here to varying extents.

The success of *The Lord of the Rings* suggests a credible scenario in which a growing proportion of film financing and film production activity results from New Zealand developed projects. To

¹⁰ This dates back to, for instance the famous Austrian theorist, Schumpeter who was publishing in the 1930s. More recent thinking is summarised and discussed in a New Zealand policy context in Harper, (1994).

examine the possible significance of this, we contrast two hypothetical projects, with the same production budget, but one developed here, the other developed overseas.

This is a simplified presentation intended to indicate the possible different effects, on the New Zealand film industry, of two otherwise like productions.

Table 6: New Zealand-developed films

Effects on New Zealand film industry via:	Project A (developed in New Zealand)	Project B (developed offshore)
Intellectual property	Local people have an ownership interest in the project's intellectual property	Significant local ownership interest less likely
Key personnel	Influence on choice of above-the-line personnel	Minor New Zealand influence on choice of key personnel
Location	Strong influence on choice of New Zealand production location	New Zealand one among many possibilities
Production facilities	Strong preference for use of New Zealand pre-production, production, and post-production facilities	If New Zealand selected for film location, possibly indicates use of local pre-production, production, and post-production facilities
Budget	Substantial proportion of budget accrues as income to New Zealand	Possibly a smaller proportion accrues as income in New Zealand
Longer term effects	Reputation enhancement with consequent growth in demand for the services of New Zealand directors and producers and support services	Important reputation effects for New Zealand individuals working on the project, but more muted reputation effects on New Zealand industry as a whole than for Project A

Hence the two projects would have very different immediate, and lasting, effects on the New Zealand film industry. It is likely that the extent of these differences would depend on the continuing record of box office success of New Zealand developed films versus films in general. Given the high level of uncertainty about how films will perform, it would be unrealistic to expect all New Zealand developed films to perform well at the box office – though *The Lord of the Rings* has now created a sound base. But the more that do, the more that the collective reputation of the New Zealand industry will be enhanced.

5.3 Scenarios

Section 5.1 sets out two broad classes of effects from *The Lord of the Rings*. We have used these to develop the scenarios given in greater detail in Appendix A.

The possible dollar value of these two effects will comprise the following:

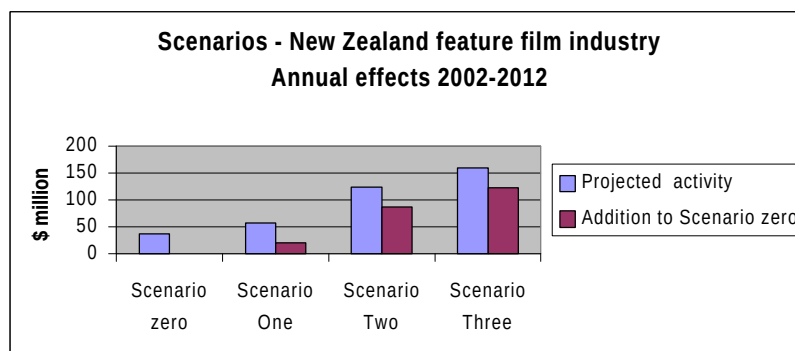
1. **Pro-active:** The additional projects developed in New Zealand by locally attached creative sponsors (or film entrepreneurs). The total New Zealand benefits will depend on:
 - The number of films that are made here each year as a result of the combined activities of the creative sponsors (entrepreneurial actions);
 - The average budget of these, and how it is distributed between the film elements;
 - The extent of local input contribution, or penetration; and
 - The duration of the effects.
2. **Reactive:** The enhanced number of international productions made here due to the increased probability of their being attracted to New Zealand. The New Zealand benefits in total will be related to:

- The number of potentially mobile films being made internationally and the proportion that come to New Zealand;
- The average budget size of these productions and allocation to ‘above-the line’ and ‘below-the-line’ components;
- The distribution of these films across ‘genres,’ to allow for differing potential local contributions;
- The eventual choice of extent of local input contribution or penetration; and
- The change in probabilities of internationally mobile films being attracted here, and the duration of these effects.

Further exploration of these ideas in detail is set out in Appendix A. There we consider four scenarios, each driven by a different set of assumptions about the way *The Lord of the Rings* works through into the New Zealand film industry.

The results are able to be displayed in graphical form, as below.

Figure 5 Economic scenarios



Source: NZIER

6. NEW ZEALAND FILM PRODUCTION COMPONENTS

6.1 Film production and post-production

Interviews with key personnel in selected parts of the New Zealand film production sector provided many of the key insights into the workings of the film industry labour market, the dynamics of the film industry, and the specific effects of *The Lord of the Rings* on their sub-sectors.

These interviews included personnel from:

1. Three Foot Six Limited

This production company is contracted to New Line Cinema to produce *The Lord of the Rings* Trilogy. Three Foot Six Ltd was established specifically to undertake this production.

2. Weta Digital

A specialist production facility in Miramar, Wellington, principally responsible for digitally created special effects for the Trilogy.

“Recruitment was not easy. The pay levels were below US standards and the sliding NZ dollar made things worse. (The new attitude to immigration and particularly to the partner’s work visas have been very helpful.)

But essentially, people came to work on The Lord of the Rings. The project itself was the big draw. The kind of professionals that a studio like Weta Digital seeks, are attracted by their interest in the work. Their employment pattern is to dive headfirst into an all-consuming project and work — virtually to the exclusion of all else — and at the close take a period out of the workforce doing something different. They will then seek out the next big thing.

The primary need is for the right kind of people. Many of the skilled people had to come in from abroad, but some of these are New Zealanders who have been attracted back.

The need to assist the staff with their international movement has led — after some teething problems — to a positive relationship being established with NZIS — the immigration authorities. This followed a discussion with NZIS and a seminar on site to demonstrate the needs. The attitude change was seen by Weta Digital as a result of an effort to build an ongoing partnership.

As a result of The Lord of the Rings, significant numbers are looking at residency — perhaps 12 done and 35 in prospect. They like the lifestyle — but are dedicated to film and will not be likely to disperse into the IT or graphics industry generally if there is no film work here.

New Zealand’s size and isolation makes it hard to get continuity of work here — artists in this industry need to be nomadic.

An interesting aspect of the Weta Digital growth has been the effects they have had — and continue to have — on their suppliers. Every vendor will have learned a lot from the sheer scale of the installation at Weta Digital. They will have gained skills and experience. Suppliers have included air-conditioning, travel bookings, and specialist data processing.

Their ability to keep up is driven partly by constant learning on the job, and partly by better gear and thus ability to create different effects easily.

This all suggests that the key capability here is the managerial talent that can:

- Organise the rapid build up and manage the run downs;
- Keep the team performing as the pressure goes on;

- *Cope with the needs of the creative side and hard deadlines while still running a coherent shop;*
- *Stay in touch with the people who are needed for the next stage, or the next project;*
- *Track the budgets and organise the capital spend; and keep the shop at the forefront of creativity in its own fields.”*

3. Weta Physical

A specialist production facility in Miramar, Wellington, which designs and creates make-up and prosthetics for actors, weaponry, armour, creatures, miniatures, and model effects.

4. The Film Unit

A facility which offers a range of post-production services including processing of negative, telecine (converting photographic images to digital form) and sound post-production.

“It aims to be ‘state of the art’ rather than a discount operation. This involves ongoing reinvestment in equipment, which only stays ‘leading edge’ for 12 to 24 months. However, the constraint on realising this ambition, is the ability to generate the necessary income to fund the required investment.

So two points follow:

- *in practical terms, such investments have to be project linked; and*
- *given the size of the New Zealand industry, international work has to be secured.*

Challenges

- *Overcoming the ‘tyranny of distance’ in networking with the international film industry, this includes the ability to access existing skills and ideas, and also to develop the new ones. There is also the problem of being in the right location to capitalise on emerging credibility in terms of technical skills by linking these into packages with the critical development, funding, and distribution aspects for substantial projects.*
- *Managing the exchange rate risk when most equipment and materials are imported, but customers want to have fixed price deals.*
- *Avoiding marketing New Zealand as the ‘2 dollar shop’ of film production, when TFU is carefully and deliberately equipped and positioned to be a quality operation.”*

5. The Wardrobe Department

The Costume Designer started on the project in April 1999, and the costume department was established in July of that year. Filming ran from October 1999 to December 2000. Over the whole *The Lord of the Rings* project over 15,000 wardrobe items were made, in well under 2 years. Most of the wardrobe department of about 50 were women. Of these, the core of about 10 had worked on *Hercules* and *Xena*. Another sub-group had significant film industry experience, and others had theatrical or some film experience. The balance came from manufacturing backgrounds.

6. The Accounts Department

In parallel with establishing dynamic budget processes, the head of this department (the Financial Controller, who joined the project in September 1998) established systems for the accounting function. *The Lord of the Rings* was seen as a ‘tight’ project as every significant item was subject to a purchase order, signed off by the Accounts Department and every piece of accounting information, such as payments, receipts etc., passed through the Accounts Department. This gave a systematic basis to the tracking and prediction of costs and budgets, that is at the heart of the accounting task. The practice was to process every bit of information as quickly as possible so that ‘real time’ reports and checks against budgets could be run at any time.

“At the height of activity there were 13 people in the Department almost all of whom were trained on the job for their roles (two were brought over from the Lost in Space feature film production). Most of those recruited had no previous film industry experience and only a few had accounting experience (one chartered accountant). They were chosen on the basis of their work attitude, flexibility, and stamina, rather than on specific relevant experience.”

7. The Stunt Department

This comprised several specialist co-ordinators brought in from the US and UK, but most of the Department, over 70 people at its peak, were New Zealanders.

“The stunt industry in New Zealand is still relatively immature. Employment is basically ‘freelance’ usually involving relatively short assignments. So The Lord of the Rings was unusual in providing sustained contracts for up to 2 years for many of the stunt team, with intensive involvement over much of that production period. This provided a unique chance for on the job learning and skill development, as well as a ‘pressure-cooker’ experience build-up.”

8. The Art Department

The Art Department Manager started work on *The Lord of the Rings* in February 1999, and her full-time involvement continued to about August 2001. Principal responsibilities were management and deployment of resources, and timing and co-ordination of different areas, involving controlling budgets, budgeting sets, reporting on progress against budget, managing people.

The Art Department is responsible for set construction and all the props that physically go on sets.

At the start, it was expected to employ a maximum of 200 people, but the actual peak was about 400 people across 11 studios.

7. FILM INDUSTRY CAPACITY IN NEW ZEALAND

7.1 Definition/discussion/determinants

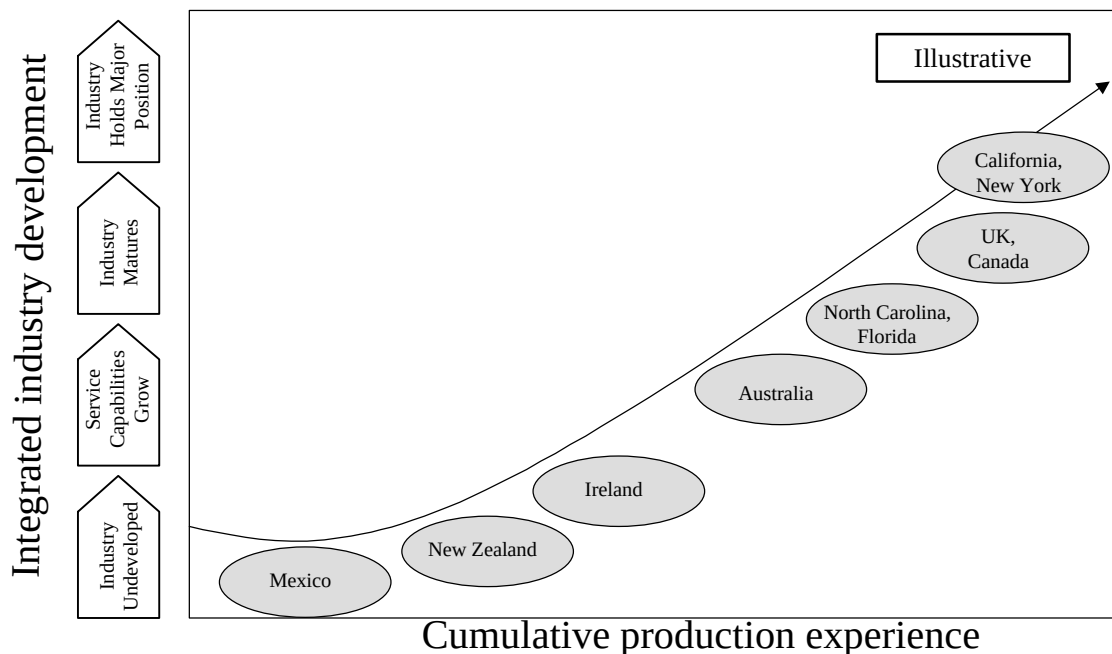
By ‘capacity’ we mean the actual and perceived ability of the film making resources (including people) in New Zealand to deliver to international standards in a cost-effective way.

International perceptions are important because the majority of feature films will continue to be ‘bankrolled’ by international investors, including particularly the major overseas studios, especially those in the United States.

The perception in 1998/99 of New Zealand as a production location, relative to North American and other locations, is indicated by the following figure.

Given the timing of the study, we assume that the illustrative position of New Zealand on this chart was not influenced to any extent by *The Lord of the Rings* project.

Figure 6: Industry development and experience



Source: The Monitor Report (1999, p.25.)

If the capacity, or competitiveness of the New Zealand industry has been enhanced by *The Lord of the Rings*, then so will the present value of its future earnings. This will be some function of the international **quantity** of proposals for films, the **probability** of their being made in New Zealand, their average budget **value**, and **penetration** of New Zealand production inputs.

In order to assess this, we need to understand the factors that are considered in choosing a location for producing a film, and the usual processes that underly the decisions.

“Looking at it from the perspective of studios or producers considering New Zealand as a production location, with no prior experience of the country, the sort of questions they would ask are:

- *What depth is there in crew available — is there a single A team and then a B team — if, for example, two feature films were to be scheduled at the same time?*
- *What about the rest of the film industry infrastructure?*
- *Is there enough movie-related equipment (other than highly specialised things like elaborate cameras that always have to be shipped from the major international centres of movie making)?*
- *Can a wide range of likely requisites be hired, such as marquees?*
- *Are the vital general skills available, such as carpenters for set construction, electricians, and so on?*
- *How well do things work, e.g. in a logistical sense, what if you have to move a lot of people around, is there the transport, reasonable accommodation, repair infrastructure?*
- *What's the cost structure: wages, on-costs, accommodation, airfares, and the exchange rate?*
- *What's the lifestyle like for 'foreign' personnel who have to relocate to New Zealand for extended periods?*
- *What about the tyranny of distance, flying to and from Los Angeles? (Maybe concerns from people about being able to get back to the US)."*

As outlined above, these inquiries had the usual two sides: how do things work, including what is available on the spot, and what do they cost?

Because feature films are both high cost and high risk, any elements of the overall 'package' likely to significantly impinge on the return/risk ratio will be important. In marketing terms, films combine creative, technical, and fashion elements, so quality signalling (e.g. through the reputations of associated key individuals — 'from the people who brought you ...') is critical.

8. PRODUCTION CAPACITY

Capacity, in terms of ‘soft’ and ‘hard’ infrastructure, is the result of a learning and building process. Skills are enhanced by experience; physical capital largely accrues over time.

So, many of those employed on *The Lord of the Rings* had worked on previous Peter Jackson films, other New Zealand feature films, and/or on long-running locally made television productions, *Hercules* and *Xena*, and/or the recent internationally mobile production, shot around Queenstown in 1999/2000, *Vertical Limit*, and television commercials. These, and other productions, going back to the early-1970s, established an experience and skills base, with some physical capacity. It also contributed to raising New Zealand’s profile as a proven and successful production centre.

“The major effect of The Lord of the Rings was to give a substantial pool of traditionally ‘freelance’ type people sustained employment on an intensive project, in contact with lots of creative talent. They got experience in 2 years, that would otherwise have taken 5–7 years to accumulate.”

Each of the production components discussed earlier involve different operations with different skills requirements, timing and duration of involvement, and *modus operandi*. However, in discussing with this diverse set of individuals, the implications of *The Lord of the Rings* for them, a number of common themes emerged:

- All had previous experience, in feature films, TV series, or TV commercials, and were attracted by the specifics of the project. This made them keen to be involved, and to make the commitment step, without knowing quite what it would mean to them in the longer run.
- For the offshore people, including many of the cast, the size of the project meant they had to be in New Zealand for months or even years. They were all accustomed to working in different locations around the world, but at the outset, New Zealand was an unknown quantity. Most of the production crew interviewed had found being here a positive experience, and did not feel they had to sacrifice much to be here.
- The size and duration of the project meant that many individuals gained the benefits of long-term contracts and associated experience, that would rarely be available on a single film project.
- For some this meant learning new skills. Most gained a quantum leap in project management experience, and with that, vastly increased confidence to take on new projects.
- For several, making the step up would have been difficult without prior experience of professionally made projects including *Heavenly Creatures*, *Hercules* and *Xena*.
- Such experience and general background made it relatively easy for them to pull in ‘core teams’ from within New Zealand.
- Almost all were self-employed ‘freelance’ operators. Working on *The Lord of the Rings* took them out of the market for other work for an extended period, but any such trade-offs were judged worth making.
- Involvement with the project offered the possibility of significantly enhanced profiles in the international film production industry.
- The size of the project created gaps and opportunities elsewhere in the New Zealand film industry.

“Film-making has been an international endeavour for many years, and being in New Zealand doesn’t mean I’m out of the loop. In my time I’ve ‘crossed a lot of paths’ and the phone is still the connection of choice for my contacts.”

I can work on overseas projects as a departmental head, where I would be able to take a few key people with me. This sort of deal is more common on 'runaway' films in third countries, where it is possible and cheaper to bring in, for instance, New Zealand crews to do things. In a film shot recently in Italy the New Zealand director/producer brought in a New Zealand lighting crew.

They were chosen on the basis of their work attitude, flexibility, and stamina, rather than on specific relevant experience. This paralleled the approach in other parts of The Lord of the Rings workforce. A number of internal 'schools' were run by experts to get large numbers of inexperienced people up to speed with a particular 'craft.'"

9. INTERNATIONALLY MOBILE FILMS

9.1 Overview

The major film studios, and the largest audience, are still based in the USA. But, as in many industries, economic forces, including changing technology have led to globalisation. From a US perspective, as mentioned, this phenomenon is known as the ‘runaway’ production. We use the more neutral label of *internationally mobile* films here.

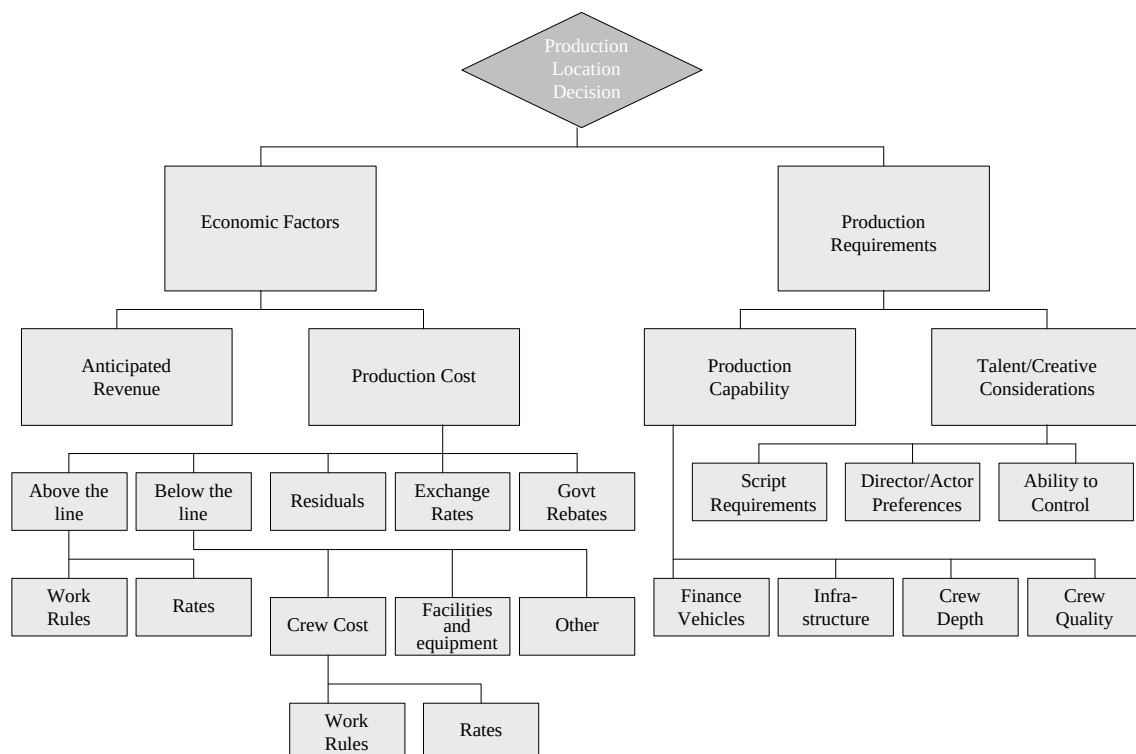
It is important to understand the factors that contribute to international migration of film-related activity, in order to properly address one of the core questions in our brief: what effect will *The Lord of the Rings* have on the probability of New Zealand being the location for future productions, and on their average size?

The analyses that have been done divide the internationally mobile into ‘creative’ and ‘economic’. They encompass:

- Feature films produced for US theatrical release
- Direct to video productions
- Movies for television, movies of the week, telefilms
- Series for television.

The following figure from the Monitor Report (1999) is intended to outline the decision making components in film location.

Figure 7: Production location decisions



Source: The Monitor Report (1999, p.18)

As illustrated in the above figure, the process of determining a production location involves balancing a set of complex economic factors against an equally complex set of production

requirements. The exact trade-off is also determined by the philosophy with regard to cost sensitivity and creative values of the company with which a producer is affiliated. The needs of the financier(s) also determine the importance placed on each set of factors.

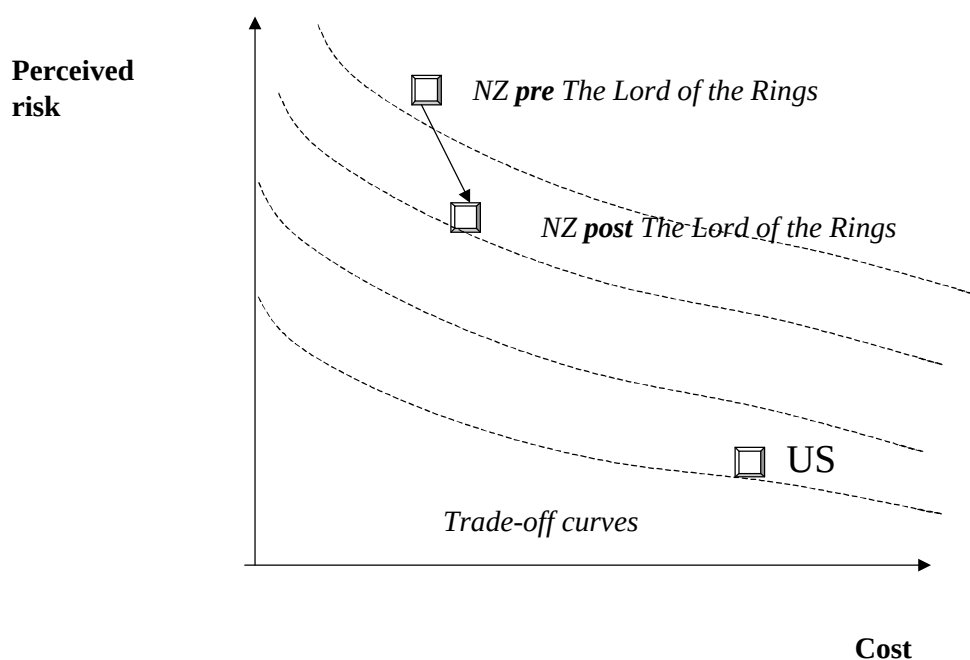
Several types of costs must be considered. 'Above-the-line' costs (such as principal actors, producers, and directors) can represent a large portion of the total production costs but in many cases much of the 'above-the-line' cost is predetermined by outside entities such as financiers who require certain kinds of talent with box office value. There is often considerably more flexibility in how 'below-the-line' costs (such as supporting actors, production personnel) are selected.

Figure 8 is an attempt to turn the general impression given by Figure 6 into a much more specific set of dimensions which would relate fairly directly to decisions about film production location.

The elements are:

- A general context in which decision-makers are trading off the critical factors of overall location related (below the line) **cost** (horizontal axis) and **perceived commercial risk** (vertical axis) in choosing production locations.
- Within that 'space' there are a series of trade-off curves (the dotted lines) for different styles or genre. The way these curves are drawn indicates that for any selected cost, less risk is preferred to more, across the cost spectrum.
- The USA is pictured as a relatively high cost, but low risk benchmark, for most genre. New Zealand on the other hand, is pictured as a relatively high risk but lower cost locale.
- The effect of *The Lord of the Rings* is to reduce significantly the perceived risk of basing production in New Zealand.
- Actual New Zealand dollar costs of producing here may have increased (e.g. because skill levels have risen).
- Overall, New Zealand's competitiveness increases significantly because the reduction in perceived risk, and enhanced perceptions of quality, more than outweigh any adverse cost effects.

Figure 8: New Zealand as a location for internationally mobile productions

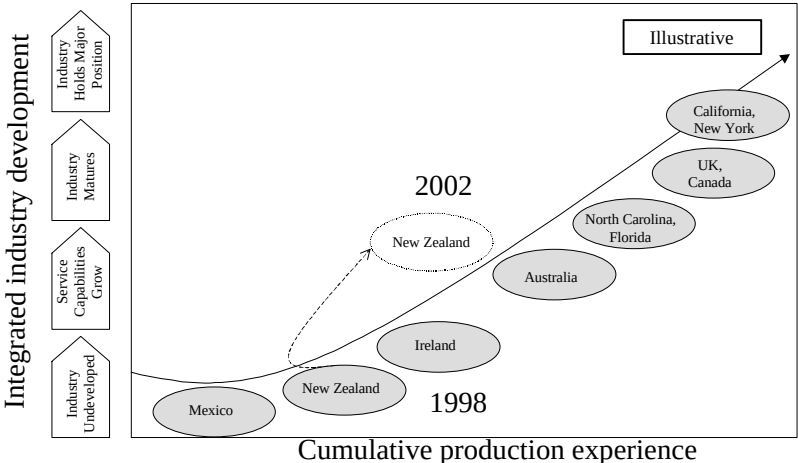


Source: NZIER

We can also illustrate the effects of *The Lord of the Rings* using the earlier portrayal (Figure 6) of film industry development. Figure 9 suggests how New Zealand's relative position and standing may have changed over recent years. Note that this is as much a subjective judgement as an objective one.

In this diagram, the New Zealand film industry is shown to have made a significant move along both axes, consistent with the discussion in previous sections. Both service capability (vertical axis) and cumulative production experience (horizontal axis) have been boosted by *The Lord of the Rings*.

Figure 9: Development of the New Zealand film industry 1998 2002



Source: Based on Monitor Report 1999, p.25.

10. OTHER SPIN-OFFS

10.1 Non-film spin-offs

We can consider non-film spin-offs within a simple ‘bullseye’ model, which reflects the overlap between:

1. The production of *The Lord of the Rings* and the inputs provided or used by other (non-film) industries;
 2. The screening of *The Lord of the Rings* and the nature of the market — existing or generated.
- Near the centre of the ‘bullseye’ are those activities or markets where there is considerable overlap.

Input examples include, specialised telecommunication links¹¹, and mobile catering services. These are important supporting activities for film, and in turn, *The Lord of the Rings* would have represented a significant proportion of their demand in some periods.

“I see myself as an enthusiast for New Zealand, and I could not see any real reason why a studio should hesitate to come to New Zealand. For instance, I see the geography as varied and unknown, and there is a very supportive environment where the full range of subcontractors (catering, and so on) have been tuned up for the needs of movies.”

Further out are government services (e.g. resource consents, immigration procedures, and advice on health and safety rules) which while important, were a less significant part of the supporting activity.

Market examples include tourism and merchandising. The initial film has done extremely well. So the three films are likely to reach huge audiences. Some proportion of these people will be interested enough to visit the various sites where film-shoots took place, or to purchase the associated miniatures and other merchandise. The effect on inbound tourism could be significant for New Zealand.

Spin-offs can be negative, for some parts of society, as well as positive. For example, there have been complaints that *The Lord of the Rings* :

- Pushed up house prices in Wellington’s eastern suburbs;
- Pushed up labour costs;
- Had adverse environmental effects in some locations where the ecosystem is fragile;¹²
- Caused parking congestion in Miramar.

Most of such effects would be concentrated in short periods and on fairly small areas, and are not unambiguously negative — one person's house price increases the wealth of the owner.

However, to the extent that one of the effects of *The Lord of the Rings* is to increase the probability of feature films being made in New Zealand, the cumulative effect cannot be ignored.

In a major project such as *The Lord of the Rings* congestion effects may be more evident than for other film projects. For example, rents and house prices in Miramar and Seatoun may have been influenced when significant numbers of offshore workers were flooding into the area. But it would be hard to prove that the scale of such effects was any greater than seen from time to time in housing markets, for all sorts of other reasons.

¹¹ These were developed specifically for the project – to ensure that communication was available for the project, no matter where the principles went on location. The innovation that was produced to serve this need is now being marketed more widely.

¹² Website: <http://www.scoop.co.nz/archive/scoop/stories/b8/b7/199911122112.913eee33.html>

10.1.1 Tourism

Walker (2001) provides a valuable review of film related tourism overseas and possible implications for New Zealand.

His thesis notes that the notion of an ‘advertisement’ is changing — messages are being embedded in films and TV programmes. Feature films may be seen as a valuable vehicle for specific ‘product placement’. Similarly, they can also be a catalyst for destination-specific tourism, when the film’s messages are memorable and durable. Further, individual films are likely to reach much larger audiences than specifically targeted tourism promotion.

But to-date, most of the research on film-induced tourism has concentrated on examples in the US, and to a lesser extent on Canada, the UK, and Australia.

His analysis includes:

- Film-related motivational factors. These are diverse and may include scenery, events in the film, images associated with actors.
- Displacement effects on tourism: it is not always clear whether film-induced motivational factors are sufficient, in themselves, to change the country of destination, or just alter the localities visited, within a given country.

A practical question is how precisely film audiences can identify the location of particular scenes? The evidence seems to be that those interested enough will find out (e.g. from film credits) or will be guided by entrepreneurial activity related to the particular locations. The ability to recognise promotional opportunities and to act in anticipation of demand, is critical to maximising benefits.

In the US setting, for example, it has been possible to identify and measure destination-specific film-induced tourism effects. Very little empirical work of this kind has been done in New Zealand. Various films — *Off the Edge*, *The Piano*, *Vertical Limit*, for example — are mentioned in discussion of possible film-induced tourism effects in New Zealand.

Some methodology that the New Zealand Tourism board has developed allows an idea to be gained of the “exposure” that New Zealand has achieved, as a product, through the audiences who have already been to see the first film. The approach is to see the film as the equivalent of a promotional piece and work out what the exposure would have cost to access commercially. Based on the attendances (including estimated US audiences based on ticket sales) through to 12 March, and making a range of assumptions, it is possible to estimate¹³ the equivalent worth of the exposure as reaching groups that would otherwise cost in excess of \$US41million.

While there is little firm basis yet for assessing the magnitude of such effects on flows of people and their associated spending in New Zealand, resulting from *The Lord of the Rings*, the success of the film, and the fact that there is such an extensive and established group of enthusiasts for Tolkien, suggests that there will be ongoing interest in the films and their making. A further indication has been the general interest in the project including its web-sites.

Taken together these suggest that tourism spin-offs for New Zealand could be significant. It will need to be backed up, however. As mentioned above, associated destination related entrepreneurial activity (such as tour organisation) will have to complement the possible motivation factors, to allow the enthusiasts to carry through on their inclinations.

¹³ See detail in Appendix G

10.2 Merchandising

A significant 'bankable' benefit, beyond the actual production work, has been in developing a place in the merchandising business.

Weta Physical is in a joint venture with a US company to make collectibles based on images and objects associated with the work they have done for the film. Weta Physical produces the masters based on *The Lord of the Rings* characters, objects and events; the figures are then mass produced in China, and distributed by the American partner. Another, similar deal has been done with the Henson Group to produce a series of Muppet-based collectibles.

One lesson from this is that looking to merchandising rights would be a major consideration in the way Weta Physical pitches for feature film work in future.

II. INITIAL CONCLUSIONS

II.1 Overview

The lasting effects of *The Lord of the Rings* will not be observable for several years, so conclusions about these effects now are to a considerable degree speculative. However, in the spirit of the ‘scoping’ nature of this study we have drawn a set of initial conclusions. These can be tested over time, as more information becomes available.

We have to start with the observation that *The Lord of the Rings* is one of the largest film projects ever — encompassing three significant films. As such, it has had unique effects, which extend beyond the mere scale of its transitional effects — which are significant — and it will leave a unique ‘footprint’ behind when it is over.

It has been written, pre-produced, filmed and post-produced here because of a number of very specific factors, but mainly because New Zealand has Peter Jackson, a creative sponsor and entrepreneur with a total commitment to the New Zealand-based film industry. He has the necessary international credibility in the film industry, and has therefore been able to develop a major project here.

In addition, he has been prepared to develop, and invest in support services (such as production and post-production facilities), to enable the film to be largely completed within New Zealand.

II.2 Transitional effects

The following data shows the transitional spending and employment effects of *The Lord of the Rings*, from the beginning of pre-production (mid 1998) to early March 2002. The grand total of costs associated with the project for that period was just under NZ\$480 million.

The following table shows:

- The division of these costs between ‘above-the-line’ and ‘below-the-line.’
- Divisions between local and foreign currency costs.¹⁴
- Major individual components of production and post-production costs.

¹⁴ The ‘local currency’ figures include costs attributable to individuals who receive payment in foreign currencies but are New Zealand residents for tax purposes.

Table 7: The Lord of the Rings costs, mid-1998 March 2002

All costs	Local currency NZ\$million	Foreign currency NZ\$million	Grand Total NZ\$million
Above- the-Line	45.8	58.3	104.1
Production			
Construction	25.1	0.4	25.5
Electrics	5.6	5.6	11.2
Transportation	12.5	0.5	13.0
Locations	31.3	0.8	32.1
Digital effects	99.0	11.6	110.6
Second unit	10.1	3.8	13.9
Main miniatures	8.5	6.5	15.0
Miniatures/creatures	28.0		28.0
Other categories	74.8	17.2	92.0
Total Production	294.9	46.4	341.3
Post-Production			
Editorial	7.0	3.7	10.7
Music	0.8	8.8	9.6
Other	1.2	1.5	2.7
Total Post-Production	9.0	14.0	23.0
Total other	3.0	7.1	10.1
Total Below-the-Line	306.9	67.5	374.4
Grand Total	352.7	125.8	478.5

Source: Three Foot Six Limited

The next table shows an equivalent breakdown of the labour costs included in the above figures.

Table 8: The Lord of the Rings labour costs, mid-1998 March 2002

Labour Costs	Local currency NZ\$million	Foreign currency NZ\$million	Grand Total NZ\$million
Above- the-Line	35.2	39.9	75.1
Production	148.6	21.1	169.7
Post-Production	3.9	4.0	7.9
Total Below-The- Line	152.5	25.1	177.6
Grand Total	187.7	65.0	252.7

Source: Three Foot Six Limited

On the employment side, estimated total person days from 1997 to early 2002 were 688,000.¹⁵ This is equivalent to about 3,200 person years, or, say, an average of 800 full-time equivalents per year for 4 years. During peak periods of production the crew count would have risen to around 1,500 per week¹⁶. In the post-production period, the headcount will drop to about 350 per week.

Comparative data for the New Zealand screen production industry is shown in Table 9.

Table 9: Screen production industry trends

March years	1999	2000	2001
Production financing:			
Feature films	\$16m	\$231m	\$308m
New Zealand total (1)	\$307m	\$497m	\$572m
Employment:			
New Zealand film and video production (2)	2,240	2,570	2,860

Notes: (1) Films, television and other. (2) Full-time equivalents as at February.

Source: Colmar Brunton, Statistics New Zealand (2001) pp.6, 19.

To date, the project has used the services of about 5,000 vendors, most of them in New Zealand.

11.3 Lasting effects

Beyond these large transitory effects, there are other effects that we expect to have a significant duration.

We have identified seven main categories of lasting effects:

1. Raising the international profile of the New Zealand film writing, directing, production and post-production industry, and sections, and talented individuals within that industry.
2. Broadbased up-skilling of people in the New Zealand screen production industry, at both technical and management levels.
3. Setting the foundation of New Zealand-based creative entrepreneurship, centred on the film industry.
4. Encouraging a significant attitudinal change amongst New Zealand writers, directors and producers towards larger projects and more determined pursuit of investors.
5. Broadening and deepening film-related infrastructure, such as buildings, equipment, and telecommunications links, and contributing to a more 'user friendly' regulatory environment.
6. Enhancing 'Brand New Zealand'. For example, opening up additional New Zealand-based tourism to cater to those who saw the films and want to visit the locations.
7. Spin-off potential, such as a merchandising/miniatures industry.

¹⁵ Estimates provided by Three Foot Six Limited

¹⁶ This number excludes extras but includes direct support service labour such as catering.

Each of these is reviewed briefly below.

11.3.1 International profile

In assessing the lasting effects of *The Lord of the Rings*, and trying to quantify them, consideration of international perceptions and relationships is critical. This is because overseas sources represent the major potential for growth in funding of New Zealand-based films. This is regardless of whether they are internationally mobile or New Zealand films about local themes.

We have concluded that by ‘showcasing’ a number of aspects of the New Zealand film production sector to the international film industry and audiences, there should be an ongoing increase in the probability of international feature films being produced here. This will come about because of positive changes in overseas perceptions about the capability of the local industry, and thus lower the potential risk attached to choosing New Zealand as a location. This provides a better environment for the local talent to seek the type of overseas support (including financial) that seems to be vital for the more substantial productions.

At the same time, there will have been a boost in the way New Zealand is seen as a location in which creative projects thrive. This will stem from the perception that productions that come here for ‘below-the-line’ reasons, can be given the best possible support, in terms of flexible crews and support facilities, and technical solutions to a variety of film related problems.

11.3.2 Upskilling

The project represents a major boost to the development of the film industry here; an industry that has been evolving in New Zealand since about the 1970s. The size and duration of *The Lord of the Rings* project mean that many of those involved gained ‘accelerated’ experience, during the project. So the upshot was that they have become seasoned in their occupational roles in ways that would have otherwise taken many years to accumulate, even in Hollywood, and that a pool of experienced and capable people has come into being. We were told, moreover, that the experience of working alongside some of the ‘greats’ from the international film industry, has given locals an understanding of the way films are made internationally, while allowing the development of a New Zealand style.

The immediate effect has been to significantly increase the collective skills — both managerial and creative — and experience, of New Zealand-based people in film and related industries. Our judgement is that the boost to this ‘soft infrastructure’ will have the largest ongoing influence on the evolution of the film industry here. It clearly offers local support for the efforts of creative entrepreneurs.

It will also significantly increase the attractiveness of New Zealand as a location. In effect, the ability of the local industry to both credibly bid into, and actually carry through significant support activities for the development of the ‘below-the-line’ based activities in New Zealand has been enhanced.

11.3.3 Creative entrepreneurship

The whole exercise will have engendered, even if merely by its existence, confidence in New Zealand writers, directors, producers (‘above-the-line’ contributors). It provides the example of a New Zealand entrepreneur working up, and then producing, a major project in this country. The success it is having should encourage this type of strategy. It may be that other New Zealand film industry participants shift their attitude, or try new strategies, to enhance local creative and production prospects.

This creates a credible scenario in which a growing share of film production activity in New Zealand is based on locally developed, and driven, projects. This in turn, suggests a degree of self-determination for the local industry, which would not arise if it had remained totally dependent on the small amount of domestic funding plus some proportion of internationally

mobile films (which are, by their nature, driven by others). The latter group will, however, remain a very important component of the local scene, for example, in contributing to the throughput necessary to sustain the economic viability of the industry.

11.3.4 Attitude changes

The effects on attitudes in New Zealand will go beyond those discussed above relating to the enhanced confidence of 'above-the-line' contributors. It will encompass the full gambit of specialists and skilled staff.

We were surprised at the effect that the successful contributions made to the project had on the confidence of the people involved — a variety of departmental managers for example. Even those with significant prior film experience were emphatic about the expansion in their own perceived — particularly managerial — competence. The number of Oscar nominations received by *The Lord of the Rings* underlines the recognition the project has brought to a number of talented local individuals.

11.3.5 Infrastructure

Another effect has been through investment in specialist production facilities including commercial structures and specialised up-to-date equipment. So, major investment occurred at the Film Unit, as well as at Weta Digital and in the area of film editing. An interesting side effect is the improved level of equipment installed by competitors of the outfits actually associated with *The Lord of the Rings* (they had to keep up to avoid losing all their market). It also means that the people working with the machinery (including the other New Zealand film-makers) are on top of the industry standards; and that the next generation of equipment is installed into facilities which are appropriate to the technology.

Another specific development was in response to the need for sophisticated telecommunication devices to allow the principles to be in touch wherever they were in the country — including on location in the backblocks of New Zealand. The reaction to such a request by their local supplier (Telecom) was to develop unique software that would allow this. The product concerned is now being seen as more widely marketable.

11.3.6 Branding New Zealand

The whole exercise, of making, and publicising the three separate films with their resulting wide circulation for the product and thus the images of the country that are contained within it, provides an opportunity for New Zealand exposure. Aside therefore, from the movie itself, this could be seen as having a general impact on the task of creating a more complete picture of this country and its prospects, to audiences around the world. Estimates of the equivalent cost of audience access provided by the film already on release suggest a significant marketing exposure.

This effect would link up with the work going on in other areas of generic New Zealand brand marketing — particularly in tourism — and could, by leaving a positive impression of the country, create opportunities for a range of New Zealand experiences or products.

11.3.7 Spin-off industries

The skills, reputation and connections acquired or exhibited through the process of producing *The Lord of the Rings* in New Zealand have, in some cases application outside the actual film industry. An interesting example is the work that Weta Physical is undertaking in producing miniatures and other memorabilia associated with the film. The quality of this work has been recognised, and they are now in collaboration with the group who have the rights to the Muppet characters to produce similar products using their creations. Prospects in this industry are basically what they make of them.

Weta Digital, still very busy with the post production work on the second and third films, are considering the possibilities of spin-offs in areas such as video games.

11.3.8 International ranking

All of these, taken together, should mean two things:

1. That the atmosphere surrounding, and thus the potential for success by New Zealand 'talent' as creative forces in the film industry has been significantly enhanced; and
2. That any future international productions in the country will be 'denser' in local content than was previously likely. That is, the need to 'import' crew to ensure a reasonable standard of film-making is substantially reduced.

We have already discussed the likely effects on creative entrepreneurship, so will deal in more detail here with the 'below-the-line' effects.

Any international production may well bring its own core management and key heads of departments, but in future, the motives will relate more to risk management and the requirements of ease of working and previous experience than to the need to have people who know what to do. The international nature of the current crew means too, that there will be natural 'referees' for the New Zealand workers, both in terms of the next international production, or in garnering work in other locations abroad.

The number of film locations, and the size of the cast and crew used, also proved that New Zealand can provide an extensive range of support services, simultaneously. These go beyond the tightly film-related to include the services associated with the administration of central and local government regulations. Such broad-based and understanding regulatory servicing ability is a favourable factor to be taken into account by international companies scouting locations.

We could speculate that the effect of *The Lord of the Rings*, on New Zealand's international image would be to improve our ranking relative to other countries, such as Ireland. Where we wind up on such a list, and whether the change will offset any contemporary shifts in relative taxes and/or other incentives commonly offered, is hard to assess.

Part of the role of a monitoring function would be to investigate films that come to New Zealand, as well as others for which New Zealand is considered but end up being produced elsewhere. It would be useful to know which other countries are considered and the critical success factors in each case.

11.4 Limits

There are real limits to the weight that can be placed on these conclusions, for the following reasons:

- Because we are mainly concerned with lasting effects, we need to make a large number of assumptions about the relationship between New Zealand's risk/return profile in the international film industry and the probability of films being made here in the future.
- There is no obvious yardstick by which to measure this risk/return profile – it is based on perceptions rather than being amenable to quantification.
- As charts in the body of the report (preceding) indicate, there is possibly some empirical relationship between the profile and probabilities, but this is difficult to disentangle from other serendipitous influences that will bear on film location decisions.
- Even if we could estimate these parameters, we would need to apply them to projections of international film activity, for example, aggregate spending, and number of new projects. Such projections will always be subject to large margins of error.

- It is very difficult to assess (in dollar terms) the size of these effects, or their duration. Such an assessment requires a whole range of assumptions including those about the success of New Zealand-based creative entrepreneurs, in launching New Zealand originated films. It also depends on film production activity world-wide, the number and size of productions that are potentially mobile (outside the US), the likely increase in New Zealand penetration, and the extent of the probability boost, mentioned above. Some indicative figures are provided in Appendix A.

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APPENDIX A: QUANTIFICATION OF FILM PROSPECTS

1. Framework

Our framework here uses the ideas developed in the body of the report. It relies on the following assumptions:

- The major market (source of demand) for New Zealand (international) film production capability will be the independently financed creative sponsors (directors/producers), and the studios and financiers who back such people.
- New Zealand has two main sources of films from this market. First, may be films originated here that will stem from New Zealand's own pool of creative sponsors. The realm of this group is not fixed, and could easily expand over time, depending on success in tapping into international risk capital, e.g. from Germany.
- Second, New Zealand is competing for some share of the 'internationally mobile' films, the number and budget value of which will be loosely related to total production activity in the US film production industry. (It will, too, of course, be strongly influenced by the dominant genres popular at any time.)
- Risk management is a major consideration for investors in deciding on, or agreeing to, film production locations.

2. Internationally mobile films

Figure 7 in the body of the report sets out the principal considerations in film location decisions, namely:

1. Economic factors: anticipated revenue (probably location independent) and production costs, plus subsidiary components.
2. Production requirements: production capability, talent/flexibility and creative considerations.

Our argument is that *The Lord of the Rings* has enhanced New Zealand's film production capability, and by changing international perceptions of this has also reduced its perceived risks as a location. We would conclude from this that the lasting effect of *The Lord of the Rings* should be to increase the:

- Expected proportion of 'internationally mobile' productions that will come to New Zealand, and/or
- Average budget values of 'typical' productions, and/or
- Average New Zealand 'penetration' of the total budgets of the films that do come here,
- Likely market reach of films produced in New Zealand.

This in turn would, over time, have significant effects on activity and employment in the New Zealand film industry. These very broad conclusions are subject to various possible refinements and qualifications, discussed below.

3. Entrepreneurial activity

As discussed in the body of the report, there is potential for *The Lord of the Rings* to create or confirm the standing of one or more New Zealand-related film entrepreneurs and creative people

into positions as creative sponsors. In other words, where their standing in Hollywood, at least among some investors, was sufficient (they were clearly 'above-the-line') that they could be the originators and subsequent drivers of major feature films.

Candidates for this position include, obviously, Peter Jackson and Richard Taylor. Barrie Osborne has made public his interest in continuing his career with a New Zealand base. Beyond these three associated with *The Lord of the Rings*, are the set of New Zealand movie makers currently established in the US or Europe, who may play the role of facilitators or primary entrepreneurs. There is also the new wave of New Zealand creators, (local script writers, producers and directors) who would be able to look at arrangements beyond the *sale of an idea*, if making serious international films here were to become accepted as a realistic and reliable prospect.

As argued throughout the report, the choice of location is not an insignificant one for investors. Neither is it one that is likely to be taken on a single factor. So the picture of the prospects we are painting, is one where the possibility of the creative sponsors being able to use their power to influence location in favour of New Zealand would be enhanced, were the New Zealand location to be seen as highly competitive.

In other words, for the New Zealand-based entrepreneurial activity, success, in the form of a track record of well-made and on-budget productions, is likely to breed further success.

4. Caveat

A major caveat attached to any conclusions, is that because of the sheer size of *The Lord of the Rings* project, the period 1999 to 2002 may not represent a realistic base from which to project spending on film production in New Zealand, or employment in the sector.

As explained in the introduction to Section 3, all the discussion about the value of the effects of *The Lord of the Rings* is based on the difference between 'with' and 'without' scenarios looking forward from 2002. It is unrealistic to assume that other projects of this scale are likely to come to New Zealand in the near future, whatever the success and influence of *The Lord of the Rings*.

5. Scenario development

Volumes

Although the concept of internationally mobile films has to be considered with care, the number of these, relative to the total number of films being made is a useful guide to the size of the potential pool in which New Zealand is competing. (Note that we are concentrating here on US developed and funded feature-films - rather than films developed elsewhere¹⁷, or TV programmes).

Recall that two types of internationally mobile films were defined in the US literature:

- 'Creative' runaways, which depart because the story takes place in a setting that cannot be easily duplicated, or for other creative reasons, and
- 'Economic' runaways which depart to achieve lower production costs. (Monitor Group, 1999, Executive Summary).

The Table indicates that over the period shown, the number of internationally mobile films nearly doubled (from 96 to 171) but that the share had fluctuated around 30% of total US developed films, rather than being on a clear trend.

¹⁷ There have been examples of films developed elsewhere brought to New Zealand for production – *The Piano* has been cited as an Australian developed, French financed film.

Table 10: US-developed feature films

	1990	1991	1992	1993	1994	1995	1996	1997	1998
US creative runaways	52	41	45	45	63	64	73	57	71
US economic runaways	44	48	43	72	93	52	128	79	100
Sub-total	96	89	88	117	156	116	201	136	171
US domestic production	223	248	263	301	324	331	432	386	363
Totals	319	337	351	418	480	447	633	522	534
Internationally mobile as proportion of total	30%	26%	25%	39%	33%	26%	32%	26%	32%

Note: Figures do not include films for television.

Source: Derived from Monitor Company Report 1999, Exhibits 2,3, & 4.

The distinction made by the Monitor Group between creative and economic motivations is, to a degree, at odds with our perception of the effects of *The Lord of the Rings* on New Zealand's place in the film world. We concluded that the effects would increase New Zealand's expected share of internationally mobile films, without being explicit about the relative weight of the contributing factors (i.e. creative or economic).

We can be a little more open here, on a judgmental basis. Our (rather high level) assessment, at this stage, is that the effect of *The Lord of the Rings* is on both sides of the equation. This is because, though it seems unlikely that *The Lord of the Rings* would have reduced production costs in New Zealand relative to the US or alternative locations,¹⁸ the perceived risks associated with production costs should have diminished. In effect, such an effect would be equivalent to a cost reduction and a risk – adjusted productivity enhancement.

In addition, we think that the proven crew quality and flexibility, the creative contributions demonstrably possible, and the variety of locations and supporting services available, mean that the perceived creative potential here has increased.

So we conclude, that overall perceived creative/quality contribution (relative to cost) would have risen significantly as a result of *The Lord of the Rings*.

As the table below indicates the number of 'economic' runaway feature films is heavily skewed toward small budget films. The Monitor Report does not include equivalent data for 'creative' runaways.

Table 11: US economic runaways

	1990	1991	1992	1993	1994	1995	1996	1997	1998
Large budget films (> \$US 25 million)	0	1	3	6	10	5	24	8	24
Small budget films (< \$US 25 million)	44	47	40	66	83	47	104	71	76
Total	44	48	43	72	93	52	128	79	100

Note: Figures do not include films for television.

Source: Derived from Monitor Company Report 1999, Exhibit 5.

¹⁸ We understand that tracking New Zealand's cost competitiveness over time is going to be undertaken as a separate exercise.

Values

We have derived the following table, which shows the estimated annual production value of all US (economic and creative) internationally mobile productions.

We infer from this data that by 1998 about 100 films per annum were classified as economic runaways, with assumed average production costs of \$US13 million.

We do not have equivalent value data for creative runaways. But for simplicity, and failing better information, we assume here that their average value is the same as for the economic runaways in the same years – it may well be that these are more valuable on average..

To make relatively well founded estimates of the future value effect of *The Lord of the Rings* we would ideally look to:

- Project (US) film production expenditure, year by year, say out to 2010, or beyond;
- Project the proportion of this which is (actually or potentially) in the creative and economics runaway categories;
- Assume some ongoing increase (as a result of *The Lord of the Rings*) in the proportion of this pool that to be produced in New Zealand.

We judge the attempt to carry through this type of exercise without significantly better data to be likely to suggest spurious accuracy.

Instead we have adopted a more exploratory approach of developing a series of alternative pictures of the way the New Zealand industry could evolve over the next few years. We note, that to compile the sort of scenarios illustrated below, involves some major assumptions, along the lines of the requirements broadly sketched above. Further, the actual effects observed will be subject to all sorts of situational specific factors, such as:

- The extent and timing of the entrepreneurial initiatives taken by the potential creative sponsors;
- The preferences of key individuals, directors, producers, executive producers, writers and actors;
- The competitive response of other film production locations to the shift in New Zealand's perceived competitiveness; and
- The mix of new film projects e.g. by genre.

But we believe that the presentation of the scenarios provides information that allows the report's readers to select their own version of the probable futures, out of the material provided.

Table 12: Estimated value of US internationally mobile feature films

\$US billion	1990	1991	1992	1993	1994	1995	1996	1997	1998
Economic runaways	0.2	0.3	0.2	0.4	0.8	0.5	1.3	0.8	1.3
Creative runaways	0.2	0.3	0.2	0.3	0.5	0.6	0.7	0.6	0.9
Total	0.4	0.6	0.4	0.7	1.3	1.1	2.0	1.4	2.2
US economic runaway films	44	48	43	72	93	52	128	79	100
Memo* (\$m):	4.5	6.3	4.7	5.6	8.6	9.6	10	10	13

Note: Figures do not include films for television.

* Inferred average production value of economic runaway feature films.

Source: Derived from Monitor Company Report 1999, Exhibit 10.

6. Scenarios

To establish the scenarios we really need a standard of comparison – a sort of ‘business as usual’ without *The Lord of the Rings*’ model. This does not exist so we have taken it to be scenario zero below.

Note that each of these ‘pictures’ is an attempt to capture a version of a possible future. We have not detailed aspects of the alternatives. In particular, we have provided estimates of average activity levels **over the next ten years or so**, without going into the precise profiles of the evolution of the growth that is inherent in many of these pictures.

In the table that gives more detail, we have used a series of estimates of the various critical elements of the activity levels. We recognise that there may be a stochastic distribution of the likely outcomes, and so have included an estimated probability of the events described occurring.

Scenario zero - More of the same: Conceptually, this represents the New Zealand film industry as it would have proceeded without any effect from the trilogy. We see this as including some probability of internationally mobile films and of limited creative sponsorship – after all, Peter Jackson did put together *The Lord of the Rings* and secure funding for it. In addition, the local industry continues to work at much the same level as was prevailing previously.

There is some difficulty in precisely determining the trends, but we have made the broad probability estimates reflected in the table below (Table – next page).

This all suggests that the expected average overall level of activity would be of the order of \$37m per year.

Scenario 1 - More of the same, but better: *Growing steadily from the existing value base.* The number of films made here increases over the numbers that would have otherwise been experienced, (scenario zero) but the effect is a slow and steady upward trend. Limited entrepreneurial activity takes place, building on the enhanced reputation effects. Local production increases gradually, too, sparked along by the general increase in film business, and some slop-overs from those working in the industry, and from the availability of equipment.

The result is an average increase in New Zealand film activity of the order of \$20m per year, relative to Scenario zero.

Scenario 2: - Quantum improvement: *An increasing New Zealand share of internationally mobile films.* The competitiveness of the New Zealand location leaps. There is also some more substantial improvement in entrepreneurial activity, as New Zealand has significantly more credibility as a location. Local activity is even higher than in the previous scenario – as the effects there are repeated, with slightly enhanced levels, as the issue of sources of financing will become a binding constraint.

The result is an average increase in New Zealand film activity of the order of \$85m per year

Scenario 3 - Substantial improvement: *A takeoff in entrepreneurial activity.* The attractiveness of New Zealand locations is seen as significant by film investors and New Zealand thereby attracts several significant internationally mobile films each year over the next few years, as a result of *The Lord of the Rings*. These are around average value for internationally mobile films, with mean about \$US15 million.

There is also substantial entrepreneurial activity, including growing involvement by people with established New Zealand affiliations. In addition, there is the possibility of new talent becoming internationally credible, buoyed by the general wave of New Zealand success. Local activity is at the level of the previous scenario, limited by the availability of funding from local sources.

This brings about an increase in New Zealand film activity averaging of the order of \$120m per year.

Table 13: New Zealand film activity scenarios

Annual figures next 10 years

Source of activity growth	Number per year	Average budget (\$NZm)	Probability	Expected annual production financing (\$NZm)
Scenario Zero				
Entrepreneurial	1	30	.25	7.5
Internationally mobile films	1	35	.5	17.5
Local	10	1.5	.8	12
Total	12	na	na	37
Scenario One				
Entrepreneurial	1	30	.6	18
Internationally mobile films	2	35	.5	35
Local	12	1.5	.8	14.4
Total	15	na	na	57.4
Increase on Scenario Zero	3	na	na	20.4
Scenario Two				
Entrepreneurial	2	30	.6	36
Internationally mobile films	4	35	.5	70
Local	14	1.5	.8	16.8
Total	20	na	na	122.8
Increase on Scenario Zero	6	na	na	85.8
Scenario Three				
Entrepreneurial	4	30	.6	72
Internationally mobile films	4	35	.5	70
Local	14	1.5	.8	16.8
Total	22	na	na	158.8
Increase on Scenario Zero	10	na	na	121.8

Note: Base data is from the Colmar Brunton survey on feature film production financing. For the 5 years to March 1999 this averaged \$19 million per annum.

Source: NZIER scenarios

Employment

It seems probable that *The Lord of the Rings* experience will result in significantly more film production activity in New Zealand, than would otherwise occur. This will increase capacity utilisation, and, depending on actual magnitudes, and the nature of the additional projects, may also require increases in capacity.

The Colmar Brunton information for recent years provides some sightings as to possible relationships between production spend and the screen production labour market in New Zealand.

Table 14: New Zealand screen production spending and employment

	1998 1999	1999 2000	2000 2001
Feature films	\$16 m	\$231 m	\$308m
Television and other	\$293m	\$266m	\$264m
Total	\$307m	\$497m	\$572m
Independent contractors	6,412	12,760	29,589
Part-time permanents	552	535	541
Full-time permanents	766	1,045	1,137

Source: Colmar Brunton (2001) pp.6, 19.

It is hard to gauge from this data what the capacity utilisation effects would be. Clearly, most independent contractors usually have a significant degree of unutilised capacity available to work in the film industry. Others, however, such as significant proportions of those working for Weta Digital, Weta Physical, or The Film Unit, had very little, if any, spare capacity for months or even years during the process of filming and finishing of the three films involved in *The Lord of the Rings*.

Hence, the effects on capacity utilisation depend not only on the number and size of projects in each year, but also on the sequencing and duration of projects. This also raises the question of the number of A teams and B teams (in quality terms) that are available in practice locally. Again, this depends on a whole range of project specifics, such as the size of each film, the genre, and the type of special effects used.

Prices

It is difficult to generalise about the price/cost effects of *The Lord of the Rings*.

We can divide the discussion into two components:

- Price effects during the various production phases of *The Lord of the Rings*: There would have been immediate effects on the prices of some things or rates for some people whose qualities or skills were in demand and/or who acquired significant valuable experience during the project.
- Lasting price effects: These would vary according to supply and demand conditions in specific labour sub-markets. One of the lasting effects would have been to increase the international marketability and mobility of many of the key production staff. This is likely to have had lasting effects both on their utilisation and remuneration expectations.

Overall, then, we would assess the lasting price and cost effects as fairly close to neutral – with the apparent increases in prices often off-set by improvements in productivity. But this assessment is very much a subjective picture.

APPENDIX B: INTERNATIONAL FILM INDUSTRY

1. Risk management and flexible specialisation

Risk management strategies¹⁹

As indicated in our stylised model, feature film-making is an expensive and risky business.

Over the years, as the industry has evolved and changed, a number of approaches to risk management have been evolved, and apply to:

- Decisions to back individual projects,
- Selection of key personnel,
- Project portfolios, and
- Project management during the production phase.

Demand for feature film production skills and resources is derived from the individual projects that emerge from key ‘entrepreneurs’ - the creators and sponsors of particular projects and the investors. Since the demise of the integrated studio system²⁰ in the 1950s, following antitrust action, the main investors have gone for a ‘portfolio’ of projects to manage their risks. So, at any given time, a series of projects will be in production, each treated as a different output, and each with its own commercial structure.

Business and social networks

Decisions to back projects financially, are closely related to decisions about who will be involved. Those ‘above-the line’ have a critical influence on investor costs and confidence. And there is a ‘pyramid’ structure for selecting key people. This operates through the hierarchy on the production side to allow those with responsibility to feel comfortable with their critical staff.

The director or producer will appoint their own key department heads, and they, in their turn, will bring the core team members into the project. Their selection criteria seem to be based strongly on previous experience – with the critical question being whether the candidate is likely to deliver under the particular conditions of the new project. This means that both quality assurance and risk management is based on a cascade of networks, revolving around successful prior experience.

The lasting influence of *The Lord of the Rings* on the New Zealand film industry will reflect in how these networks, involving New Zealanders (based in New Zealand, or elsewhere), or overseas people with New Zealand links, change and grow as a result of the project.

2. Labour market conditions

Self-employment

From the viewpoint of workers in the industry, demand for their services will appear lumpy and sporadic because it is project by project. This is reflected in the labour market which is

¹⁹ Australian Financial Review ‘Disaster movie: how film insurance died and took film investment with it.’ February 16, 2002.

²⁰ This had the investors, producers, and distributors in a continuous ‘factory line’ process, that allowed the use of many instruments, from marketing and promotion, through the ‘building’ of stars and so on, — including critically the integrated nature of the process — to mitigate risk.

dominated by self-employed people, many of whom are only working on film projects on a temporary, or part-time basis.

“Most people in the industry are self-employed. They see themselves as working in the ‘industry’ rather than for a particular firm or operator. And, in the same way that each production is a ‘project,’ the workers are taken on (and thus being paid) to achieve a specific necessary result in a specific time, as part of a complex sequence of time-sensitive work that makes up the whole. This suggests that the logic of the relationship makes it of the nature of one to be governed by a contract for service, not by a contract of service.”

Individual contracts

That is why ‘individual contracts’ for the suppliers are critical. Almost all work in the industry is urgent, because there is usually a rigid deadline (set by commitments to the film’s distributors – e.g. *The Lord of the Rings* release is going to involve the completion of 10,000 prints world wide, some with language dubbing) - but still, within the whole production process, a need for creative flexibility.

This stems from the time-bound nature of the process interacting with the need for creative imagination to be included to make the film into a distinct and demanded product in the market. It means that there will be constant uncertainty and new ideas coming into the shoot, even as it proceeds.

So the crews have to be able, as a group, to handle both:

- Changes to shooting that come from an unforeseen way of approaching an idea that was always in the plan; and
- Changes to shooting that are totally new and driven by ideas that were never in the plan.

Time pressures

For example, new scenes or schedule changes might mean that set builders think they have six weeks to build something, and end up with only five days to completion. And, the requirement for tight cost control means that it is normal to only crew-up to the minimum numbers - there is no ability to ‘carry’ people who are not directly contributing, and it is normal to have to replace anyone who moves (or is moved) on. This time pressure also suggests why such ‘moving on’ is common – there is no ‘free-board’ in the production organisation to ‘work around’ or even ‘work out’ dysfunctional associations. They must be addressed promptly to ensure that the project budget is adhered to, and that the risk of cost overruns or timetable blow out is managed.

Thus, the tightly time- and money-budgeted, project-by-project nature of the work dictates a certain type of ‘employment’ structure for the industry. It needs to be both serial and separable. People are only ‘brought in’ (that is, on the payroll) for the time needed to complete their particular contributions and then move on – possibly without any assured work alternatives in sight.

And, they must exhibit total dedication to the timetable inherent in the project, rather than to external factors like public holidays, and even weekends.

Flexibility

Flexibility is another logical requirement of this tightly managed project-based environment. The ‘contra’ to this type of work pattern is that those engaged typically earn ‘super-normal’ pay for the time they are employed. This allows them to have periods ‘resting,’ or to subsidise other activities.

The ‘creative’ side is never finally fixed in place until the last edit is made, so changing what is being done, is always in prospect. (This is a specific area where New Zealand crews were seen as

outstanding — there were no artificial demarcation rules, or refusal to pitch in and do what was necessary for the needs of the production.)

Another pattern of activity (like a widget production line plant) would require different organisation structures and alternative ‘employment’ conditions, to best match the economic and production imperatives.

3. Evolution of the international film industry²¹

Vertical disintegration and the creative sponsor

In the vertically disintegrated industry that emerged in the 1970s, most of the transactions required to produce a film take place ‘on a market’ rather than ‘within a firm.’²² The industry involves the production of lumpy products through relatively non-routinised processes, and is structurally designed to maximise potential variability of outputs, and maintain flexibility with respect to choice and contracting for the full range of inputs.

Films are initiated in many different ways. Frequently a creator will come up with an idea for a movie project, and pitch this successfully to an investor who will then arrange for the further development of the treatment, followed possibly by production. As part of this process, the creator may be just paid for the idea (which may be as small as a simple novel idea, or as extensive as a full script, story, novel or play), and that will spell the end of their involvement.

But often the creator is engaged in the development process as it proceeds. The form of this engagement can be various, from script associate through to a larger role (such as script writer) in the production.²³

Occasionally, the creator can be sufficiently well-regarded to assume an important role in the production process — possibly even producer or director. This will occur only when the individual concerned is credible, and probably has a distinguished track record. In these circumstances, the creator becomes a *creative sponsor* and will be able to exercise a degree of control over critical film decisions, such as the location for shooting. Many of the bigger names in the industry work in this way, developing their own projects, within which they will direct or produce.

New Zealand has had very few such people, but the success of *The Lord of the Rings* could cement Peter Jackson into that position. He was a creative sponsor for *The Lord of the Rings*, and for several previous film projects. Commercial success in film, as in other industries, requires not just good ideas but also the ability to carry them through in a deal-making sense. Prerequisites include an appetite for risk, persistence, and stamina.

Flexible specialisation

As vertical disintegration proceeds, firms providing services to producers become smaller and more specialised. They reduce their own risks by marketing their services to other entertainment

²¹ Refer, for example, S Christopherson and M Storper (1986) ‘The city as a studio; the world as a back lot: the impact of vertical disintegration on the location of the motion picture industry’, and Lois S Gray and Ronald L Seeber (1996) *Under the stars — essays on labor relations in arts and entertainment*. Cornell University Press.

²² This shift from an integrated ‘studio’ system to a disintegrated independent production house style, is a classical case of a shift within the Coasian paradigm. Coase’s interest was in why firms took the shape that they do. He suggested that one of the fundamental issues in this choice was the extent to which the production was sourced from an in-house supplier, or from the market. Risk was one of the major features driving the decision.

²³ Novelist John Irving wrote an interesting book about his trials turning *The Cider House Rules* into a movie script — and then eventually playing a minor role in the actual production as the station master.

services wider than the film business, including music recording and television. The regime that results has been described as ‘flexible specialisation.’

“...has evolved into present entity and built specialised problem solving techniques, together with the business disciplines necessary to survive in a narrow, price sensitive, and unpredictable marketplace — ‘a fickle, knife-edge industry’.”

An extract focusing on the US industry in the mid-1980s, but which seems (in parts) to closely represent what we observe in the New Zealand film industry currently, is as follows.

“Production subcontractors spread risk by serving a variety of clients with different final outputs. They specialise in certain generic functions (props and scenery, film editing, sound mixing etc) but are able to work on a broader range of product outputs. They are both flexible and specialised. All of these activities require constant access to new information, personal visibility, and the use of intermediaries, particularly lawyers and agents. Every actor, director, and goods or services contractor is also a deal maker, negotiating his or her way through a maze of possibilities and potential or actual contractual agreements.”

(Christopherson and Storper, 1986, p.313)

4. Internationally mobile productions and the world film industry

The days in which the international film industry was dominated by a few large vertically integrated studios (i.e. firms spanning financing, production, and distribution) based in Hollywood are long gone. Much of the risk capital for feature film making is generated in Europe and the UK, as well as the US, and production has been increasingly dispersed within the US and internationally.

New Zealand’s profile as a production location has been rising for some years, and should be given a significant boost by *The Lord of the Rings*. For much of the last 30 years, or so, the New Zealand profile has been built on a somewhat sporadic sequence of local productions, with relatively small budgets. More recently, as a result of its growing international profile, New Zealand has become an increasingly serious competitor with countries including Canada and Australia for ‘internationally mobile’ films, or what the US film lobbyists call ‘runaway’ productions.

5. US reports

The perceived erosion of the US domestic production industry by ‘imports’ is of significant concern in the US. Several major studies have been produced on the subject of these so-called runaway productions. The Monitor Company Report, June 1999, was commissioned by the Directors Guild of America (DGA) and Screen Actors Guild (SAG).²⁴ Another report, commissioned by the US House of Representatives and conducted by the International Trade Administration section of the Department of Commerce, was published in March 2001.²⁵

‘Runaway productions’ are defined as those which are developed, and are intended for initial release/exhibition, or television broadcast, in the USA, but are actually filmed in another country.

In this kind of analysis, there are two types of runaway productions — ‘creative’ runaways, which depart because the story takes place in a setting that cannot be duplicated or for other creative

²⁴ Monitor Company (1999) <http://www.sag.org/pressreleases/pr-la990623b.html>

²⁵ US Department of Commerce, March 2001, The Migration of U.S. Film and Television Production Impact of "Runaways" on Workers and Small Business in the U.S. Film Industry, p.38. <http://www.ita.doc.gov/media/filmreport.htm>

considerations, and ‘economic’ runaways, which depart to achieve lower production costs. (Monitor Company, 1999, p.2).

In our view, this distinction may be hard to draw in practice.

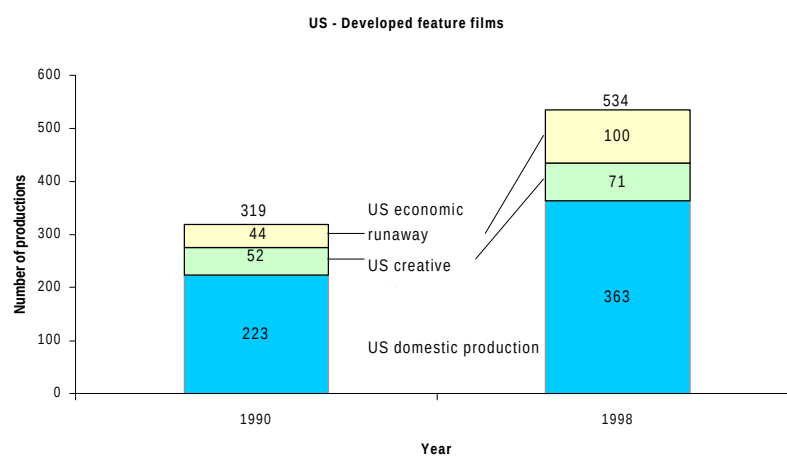
The following chart sets out summary information for the 1990s.

The Monitor Report (p.3) notes several important contextual points:

- In terms of economic impact on the industry in the US, economic runaway TV films have the largest impact (\$US2.7 billion), followed by feature films with budgets larger than \$US25 million (\$US2.4 billion), and those with budgets smaller than \$US25 million (\$US2.3 billion impact).
- It is noteworthy that feature films have such a significant impact. Conventional wisdom held that economic runaways are a television movie phenomenon, and that larger productions would remain in the US since the infrastructure required to produce them was not available abroad. This data may indicate the leading edge of a trend, with larger budget productions departing.
- The productions were leaving at a time when US domestic production had been growing, so the runaway phenomenon had gone relatively unnoticed. The number of US-developed feature productions grew at an average annual rate of 8% between 1990 and 1998. During that same period, the number of US-developed features that were made in Canada grew 17% annually.

Figure 10: US internationally mobile productions, 1990 and 1998

Numbers



Source: The Monitor Company (1999, p.2)

APPENDIX C: PLATFORM FOR FURTHER WORK

Part of our brief was to:

- Indicate the extent to which this study could provide a platform for further work relating to *The Lord of the Rings*.
- In addition, to advise whether it provides insights into how the New Zealand Film Commission might develop analytical tools to assess the contribution of film production to the economic welfare of New Zealand.

With respect to the first of these, the focus of this report on ‘lasting’ effects led us into consideration of both soft and hard infrastructure in New Zealand, changed perceptions of the capability of the New Zealand industry, and improved prospects of films being made here. In this report, we are having to speculate about much of this. In order to complete the story (i.e. to be able to report eventually on what did happen) it would be useful to set up some ongoing data collection and tracking procedures.

For example:

- At say, 6 monthly or annual intervals, tracking the employment and capacity experiences of a number of the key New Zealand-based production specialists;
- Tracking the capacity utilisation of specific production facilities, e.g. The Film Unit, or Weta Digital;
- Analysing in detail the mix of films that are made here, their budget size and motivations for choosing New Zealand as a production base;
- Monitoring film projects for which New Zealand is a potential location, but that come to fruition elsewhere. What are the critical factors that led to the choice of another location?

In each case, the analysis would have to deal with the separation of *The Lord of the Rings* effects from other influences, but such a longitudinal approach should add some important insights. This could be based on an expanded version of the existing Colmar Brunton survey.

On analytical tools, we would see the most relevant approach to assessing contribution to welfare as being cost-benefit analysis — subject to an appropriately specified ‘counterfactual.’ In this case we would be comparing economic outcomes with *The Lord of the Rings* against a hypothetical base — what might have happened without that project.

The ultimate measure of economic welfare would be some measure of GDP contribution, but the analysis could also look more broadly at some of the distribution effects. It would involve a probabilistic approach to allow for uncertainty about effects, in the future, on film production activity in New Zealand. The quantification challenges would be very much as laid out in detail in Appendix A.

APPENDIX D: THANKS

In the course of the project we trespassed on the good nature of many who had been involved with the massive production task that *The Lord of the Rings* was. Those people gave us their time and tried their best to make us understand what had happened during the production and what would happen in the future.

We owe them thanks and list many of their names below.

Interviewee	Organisation	Title
Dr Ruth Harley	NZ Film Commission	Chief Executive
Mladen Ivancic	NZ Film Commission	Deputy Chief Executive
Irene Lake	NZ Film Commission	Advisor
Barrie Osborne	Three Foot Six Limited	Producer The Lord of the Rings
Janine Abery	Three Foot Six Limited	Producer s Assistant
Elena Azuola	Three Foot Six Limited	Financial Controller, The Lord of the Rings
Jamie Selkirk	Director, Weta Digital	Co-producer, The Lord of the Rings
Richard Taylor	Weta Physical	Director
Tania Rodger	Weta Physical	Director
Andrew Smith	Weta Physical	Financial Controller
Sue Thompson	The Film Unit Limited	Chief Executive Officer
Steve Bayliss	Weta Digital	Accountant
Ngila Dickson	Independent Costume Designer	Wardrobe Department Costume Designer, The Lord of the Rings
Chris Hennah	Co-Director, Morning Star	Art Department Manager, The Lord of the Rings
Tim Hansen	Independent	Production Manager, Aidiko Insane
Kirk Maxwell	Independent	Stunt Coordinator, The Lord of the Rings
Nik Korda	Independent	Supervising unit production manager The Lord of the Rings
Zane Weiner	Independent	Unit production manager The Lord of the Rings
Robert Arlidge	Industry New Zealand	Sector specialist
Paul Voigt	Investment New Zealand	Investment Manager
Reid Walker	University of Otago, Department of Tourism	Masters Student

APPENDIX E: ABOVE- AND BELOW-THE-LINE²⁶

I. Above- and below-the-line

Above-the-line: The portion of a film's budget which covers major creative elements and personnel, i.e., those which are creatively unique and individually identifiable. These are primarily story, acquisition, screenplay rights, script development, writer, executive producer, producer, director and principal members of the cast. The phrase 'above-the-line' refers to the location on the film budget of the specific expense items/person. (from John W Cones, *Film Finance and Distribution: A Dictionary of Terms*)

Below-the-line: Film budget items relating to the technical expenses and labour (other than above-the-line) involved in producing a film, i.e., relating to mechanical, crew, extras, art, sets, camera, electrical, wardrobe, transportation, raw film stock, printing and post-production. Below-the-line personnel include the production manager, cinematographer, set designer, special effects persons, wardrobe person, and make-up artist. The phrase 'below-the-line' refers to the location of the specific expense items/person on the budget. (from John W Cones, *Film Finance and Distribution: A Dictionary of Terms*)

Crew fall into three basic types:

- The first are core production and support. This is the producer, production and technical support. These crew are long-termers and are there for the whole project.
- The second group are the main crew. This group come on over an extended period, according to the requirements of the schedule. They conduct R&D, build and run pipelines and form the main body of the production. They tend to stay until, or near to, the end of the production.
- The third group are the short-termers. These crew are brought on for short periods, typically close to delivery, to complete very specific tasks.

²⁶ Monitor Company Report, 1999, p.28.

2. Who does what?

Table 15: Breakdown of motion picture & television personnel

<u>ABOVE-THE-LINE</u>	<u>BELOW -THE-LINE</u>	-Stage & Studio	-Locations
-Writer -Executive Producer -Producer -Co-Producer -Line Producer -Associate Producer -Director -Casting Director -Talent Stars Supporting Cast Day Players Stunt Players Extras: Atmosphere (SAG or Non-union) -Travel & Living Cast/Crew Transportation Cast/Crew Lodging -Atmosphere General Atmosphere Teacher/Welfare Worker Atmosphere Cast -Fringe Benefits SAG Pension & Welfare IATSE	-Production First Assistant Director Second Assistant Dir. Second Second Assistant Director Production Accountant Production Coord./Mgr. Production Secretary Office Production Assistant (PA) Set Production Assistant (PA) Script Supervisor Assistant to the Producers & Director -Art Dept./Set Dressing Production Designer Art Director -Construction Set Construction Construction Foreman Carpenters/Painters Greensman -Set Dressing Set Dresser Set Decorator Lead Person Art Swing Gang Purchases Rentals -Prop Department Property Master Assistant Property Master Rentals Purchases Animal Handlers/ Wranglers Animals Weapons Specialist -Camera Director of Photography (DP) Camera Operator First Assistant Camera Second Assistant Camera Steadicam Operator Still Photographer Rentals -Sound Sound Mixer Boom Operator Supplies Walkie Talkies	Rental Labour Equipment -Electrical Gaffer Best Boy Electric Third Electrician Swing Electric/ Grip Day Players Generator Rental Rentals (including trucks) Expendables -Grip Key Grip Dolly Grip Best Boy Grip Third Grip Dolly/Cranes Rentals (including trucks) -Wardrobe Costume Designer Key Wardrobe Coordinator Assistant Wardrobe Cleaning Rentals -Makeup & Hairdressing Key Makeup/Hair Assistant Makeup Assistant Hair Supplies -Special Effects/ Pyrotechnics (FX) Makeup Effects Pyrotechnics -Laboratory & Film Negative Film Stock Developing Negative Polaroid Film Stills Rawstock -Food Caterer/Chef Assistant Chef Craft Service Late Meals Water and Ice -Transportation Transportation Coordinator Transportation Captain Drivers Honeywagon Driver Vehicle Rentals Picture Vehicles Water Truck	Location Manager Asst. Location Manager First Aid Police Fire Safety Security Loss & Damage Restoration Telephones/Car Phones -Editorial Editor First Assistant Editor Apprentice Editor Post Production Supervisor Film Cutting Purchases POST PRODUCTION -Music Composer (Music Editor) Music Clearance -Sound Sound (ADR/Foley/ Sound EFX/Mix) -Film/Lab Work Title & Opticals (Stock Shots) MISCELLANEOUS -Publicity Publicist -Insurance Completion Bond Workman s Compensation -Legal Lawyer

Source: International Trade Administration section, Department of Commerce (2001) The Migration of U.S. Film and Television Production Impact of "Runaways" on Workers and Small Business in the U.S. Film Industry, p.11. <http://www.ita.doc.gov/media/filmreport.htm>

APPENDIX F: THE THREE PARTS OF CREATING A MOVIE: PRE-PRODUCTION, PRODUCTION, AND POST-PRODUCTION²⁷

The process of creating film and television program is generally divided into three parts.

Pre-production generally refers to activities that are performed and paid for before shooting begins. This includes script development, set design, casting, crew selection, costume design, location selection, and preparation of a budget.

Once these preliminaries are accomplished, then the second part, **production** — the actual shooting of the film — can begin.

The third part, **post-production**, is generally considered to include film editing, colour imaging, making of soundtracks, special effects, musical scoring, titles and credits, and dubbing.

The term ‘film and television production’ covers several types of production, notably:

- Full-length feature films,
- Movies-of-the week (or made-for-television movies),
- Series television programming,
- Television commercials, and
- Music videos.

Full-length feature films: Of the film and television productions, full-length feature films (also called ‘theatrical films’ in the industry) produced by both major studios and independent companies operate on the largest scale and generally require the greatest resources. The MPAA reports that, for 1999, 441 feature films were released in the United States – 213 by the major studios, and the remaining 229 by U.S. independent producers and distributors of imported foreign movies. Based on MPAA estimates, total feature film expenditures in the United States in 1999 reached \$11 billion, with \$10 billion coming from major studios and \$1 billion from independents.

Television programming: Television production includes programming of one-hour long and half-hour long episodes, movies-of-the-week (MOWs), and miniseries (a MOW of two or more parts). All the major film studios and many independents are involved in producing television programming.

Movies-of-the-week (MOWs): Due to the rapid growth of the cable industry, the production of MOWs (or ‘made-for-television movies’) has been an area of sizeable growth in recent years. MOW production has probably been more affected by runaway production than any other type of film production because it is most sensitive to cost factors. The costs associated with filming MOWs are generally far lower than those of feature films, often more in line with those of filming series (or episodic) television programming.

Television commercials: The industry standard for television commercials is the ‘30-second spot’, although 15-second and 60-second commercials are also used. According to one industry source, the production budget for a 30-second commercial can easily approach \$500,000, and some may cost as much as \$1 million. Although some small businesses in the industry may specialise in commercials, most small businesses who work below-the-line consider television

²⁷ US Department of Commerce, March 2001, pp.15,16. <http://www.ita.doc.gov/media/filmreport.htm>

commercials to be an interim source of work that is essential to sustain their companies between feature films and MOWs.

Music videos: Record companies, mostly belonging to the Recording Industry Association of America, produce music videos. This is by far the smallest segment of the motion picture and television production market.

ii) Location choices²⁸

Comments reported in New York's BCG report:

"Cost consultants have boiled everything down to the bottom line. Budgets are set and it is up to us to meet the cost requirements."

- Commercial producer

"50% of scripts already have a star or director attached to them, and will be driven by artistic requirements and star demands [intangibles]. The 50% of scripts without talent attached will be more cost driven."

- Feature film producer

"Location decisions tend to be a trade-off between costs, script and talent...a script can drive the decisions as can A-list talent, but without those factors, cost will be the primary concern."

- Feature film producer

"Our incentive is to make shows as cheaply as possible so the studio's risk is minimised. However, there are certain things that I won't compromise on, like the energy of a location which you can't fake."

- Television producer

²⁸ US Department of Commerce, March 2001, p.38. <http://www.ita.doc.gov/media/filmreport.htm>

APPENDIX G: ESTIMATING EXPOSURE WORTH

This material was sourced from Ian Macfarlane, New Zealand Tourism Board.

He used attendance figures that were supplied by Three Foot Six Limited covering audiences up to 12 March, for the following countries:

Japan	5,499,006 (first ten days of release)
North America (US/Canada)	50,989,397 *
UK	13,771,517
Australia	4,962,725
Germany	10,663,180
South Korea	3,864,447

* This number was calculated by dividing the cumulative gross box office for North America by the average ticket price in North America.

Further assumptions were made about the following:

1. Average cinema size	500
2. Average sessions per week	35
3. Total people per cinema week	10500
4. Cost of 60 sec commercial	Varies by country. See table below.
5. Total minutes of destination coverage	22
6. Dilution factor	30%
7. Total promotion minutes	11

This allowed the calculation of the worth of the exposure:

Country	Audience	Cinema weeks	Equivalent ads	Total Ads/ cinema week	\$/ advert/ cinema week	Total value
USA	50989397	4856	15	74784	\$300	\$22,435,335
Japan	5499006	170	15	2625	\$590	\$4,758,473
UK	13771517	1312	15	20198	\$580	\$11,714,970
Australia	4962725	473	15	7279	\$125	\$909,833
Germany	1066180	102	15	1564	\$260	\$406,570
Korea	3864447	368	15	5668	\$300	\$1,700,357
Total	76,443,824	7280	92	112118		\$US41,925,538